RescueSwarm: An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

Technical Design Proposal

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***Abstract*—India recorded 5.4 million internal displacements from disasters in 2024, the highest figure in twelve years. After a flood, rescuers usually spend longer finding survivors than reaching them. Boats and ground teams search slowly and at real risk to their crews, and helicopters, though fast, are few and are mostly needed for evacuation. This paper describes RescueSwarm, a system of three autonomous aircraft that search an assigned area, detect and locate survivors, and drop a standard relief kit to each one. A single operator runs the system, and it needs no mobile network or internet connection. We present the system architecture, the sizing and geolocation error models behind the design, the simulation results we have so far, and a phased work plan. We buy parts from Indian suppliers wherever a suitable one exists, mainly because domestic lead times are about half those of imports and every week saved goes into flight testing. Nothing has flown yet, so throughout the paper we mark each figure as measured, modelled or assumed.**

***Index Terms*—unmanned aerial systems, multi-robot coordination, search and rescue, coverage path planning, RTK geolocation, edge inference, disaster response**

I. Introduction and Problem Statement

A. The problem

India has one of the highest disaster-displacement figures in the world. The Internal Displacement Monitoring Centre reports 5.4 million internal displacements in India during 2024, the highest in twelve years, and 2.4 million of them came from a single monsoon flood season in Assam [1]. The scale of the problem is well documented. How quickly displaced people can be found is much less studied.

What matters most in post-flood search and rescue is the time between the start of an event and the moment a survivor is both found and reached. Existing national capability is much better at the reaching than the finding. Boat teams and ground parties get to survivors reliably once they know where to go, but they search slowly, at considerable risk to themselves, and they cover debris fields and rooftops poorly. Helicopters search quickly, but there are few of them, they are expensive to fly, and they are usually busy with evacuation.

Finding people is therefore the bottleneck. An aircraft that could search an assigned area systematically, pick out people in cluttered flood imagery and report each one as a coordinate accurate to a few metres would let every other rescue asset be sent where it is actually needed.

B. Why several aircraft

A single multirotor that carries a useful payload and flies for fifteen minutes does not cover much ground. Coverage grows with the number of aircraft in the air, but so does the workload on the people flying them, and a system that needs one pilot per aircraft is hard to use in a disaster. The engineering problem is therefore to fly several aircraft autonomously under one operator. That means dividing up the area, allocating tasks, keeping the aircraft apart, combining their reports, and behaving sensibly when the command link is lost.

C. Sourcing and schedule

We prefer Indian suppliers mainly for schedule reasons. Our supplier survey found typical import lead times of four to six weeks, compared with two to three weeks for domestic parts. Within an eight-week flight-test window, that difference can decide whether we get one round of testing or two. For that reason we name suppliers throughout and record lead time alongside price for each part we select.

Some of our own choices work against this. To save cost, the build uses an imported autopilot and imported propulsion instead of Indian ones, and those parts have the longer lead times described above. We deal with this by ordering every long-lead item in the first month, before the software work that would normally come first. We accepted this trade of schedule

margin for cost knowingly, and the remaining risk is listed in Table X.

D. Contributions

1) A three-aircraft autonomous search-and-delivery system that one operator can run without any network, sized part by part against components available on the market.
2) A geolocation error budget that treats the accuracy of survivor coordinates as the main output of the system, and an architecture that meets it using our own RTK base station rather than an external correction service.
3) A coordination scheme that combines a fixed equal-area split of the search area, back-and-forth (boustrophedon) coverage within each part, and a simple greedy claim-and-lock rule for assigning survivors. We chose it because it is predictable and easy to verify, even though it is not optimal.
4) A way of working in which every number we report can be regenerated from the model that produced it, and in which measured, modelled and assumed values are kept clearly apart.

II. Objectives and Scope

A. Primary objectives

O1 Search an assigned area autonomously with three aircraft working together, under one operator and one ground control station.
O2 Detect people in downward-facing imagery at a resolution fine enough for reliable classification, running detection on board and combining several frames to confirm each sighting.
O3 Report each detection as a map coordinate with a circular error probable of a few metres.
O4 Drop a standard relief kit at each confirmed survivor location.
O5 Work without any mobile, internet or cloud service at any stage of the mission.

B. Out of scope

To avoid the scope being read more broadly than we intend, we list what the system is not.

- It is not for carrying people. The payload is a relief kit, and we do not claim it could grow into casualty evacuation.
- It is not certified for flight beyond visual line of sight. It flies inside a fixed geofence, and regulatory approval for more is future work.
- It does not work at night or in all weather. The camera sees visible light only. Section XII explains why we think near-infrared, rather than thermal, is the better extension for people in water.
- It does not triage survivors. It reports where they are and drops a kit, and makes no medical assessment.

III. Related Work and State of the Art

A. Coverage path planning

Dividing an area among several robots for coverage is a well-studied problem. The DARP algorithm (Divide Areas for Optimal Multi-Robot Coverage) produces equal-area, connected regions, one per robot, and is the usual reference method [2]. Within each region, the classical boustrophedon or back-and-forth pattern [3] is still the standard way to sweep a sensor with a rectangular footprint.

We use a fixed equal-area split with boustrophedon coverage inside each part instead of a full DARP implementation. This gives up some optimality in return for predictability: the same inputs always produce the same plan, which we can inspect, and our verification approach (Section VI-B) depends on that. If we later need to reallocate areas during a mission, DARP is the obvious next step.

B. Multi-robot task allocation

The Consensus-Based Bundle Algorithm (CBBA) is the standard decentralised auction method for assigning tasks among several agents [4]. Its guarantees depend on a consensus phase over a communication network. Our system has only three aircraft, each with a reliable link to one ground station, and in that setting those guarantees add little. We use a simpler greedy claim-and-lock rule with a fixed tie-break instead. It is much easier to verify, and with three aircraft its worst case is acceptable.

C. Aerial detection of people

The main difficulty is that people seen from survey altitude are very small: an adult lying down covers only a few tens of pixels. Slicing-aided hyper inference (SAHI) and similar tiling methods improve recall on small objects considerably by running the detector on overlapping crops of the full-resolution image instead of on a shrunken whole frame [5]. The standard benchmarks show the size of the problem. COCO treats anything under $32^2\,\mathrm{px}^2$ as a small object [6], and TinyPerson adds finer size bands below that because people in aerial images are often smaller still [7].

Several aerial person datasets are available for transfer learning. SARD [8] covers search and rescue on land, and VisDrone [9] covers urban aerial scenes. The closest match to our case is SeaDronesSee [10], which labels people in open water seen from a drone. Like a flood scene, it has a low-contrast background and targets that show only head and shoulders. None of these datasets covers flooded Indian terrain. The proper fix would be our own field data campaign, but that is deferred from this build (Section VIII), so detection recall remains a modelled figure and is the top entry in our risk register.

D. Airborne payload delivery

In airborne delivery, most of the error comes from the aircraft's velocity and the wind at the moment of release; the release height matters much less. Mathisen *et al.* demonstrated autonomous ballistic airdrop from a small fixed-wing UAV

TABLE I
AIR VEHICLE DESIGN POINT (PER AIRCRAFT), AS MODELLED WITH THE 6S3P PACK

Parameter	Value
Configuration	Quadrotor
Rotor diameter	18 in
Maximum take-off mass	6.36 kg
Fleet mass (3 aircraft)	19.08 kg
Regulatory fleet cap	25 kg [13]
Battery	6S3P Li-ion 21700, 292 Wh
Pack nominal voltage	21.6 V (18.0–25.2 V)
Pack mass	1 449 g
Cells per aircraft	18
Peak pack current	115 A (8.5 C) at T/W 2.0
Static thrust-to-weight	2.0
Hover endurance	15.3 min
Design mission duration	7.6 min
Survey altitude	40 m AGL
Ground sample distance	$\leq$2.0 cm/px
Survey groundspeed	8 m/s
Payload	4 kits $\times$ 200 g
Geofence radius	600 m

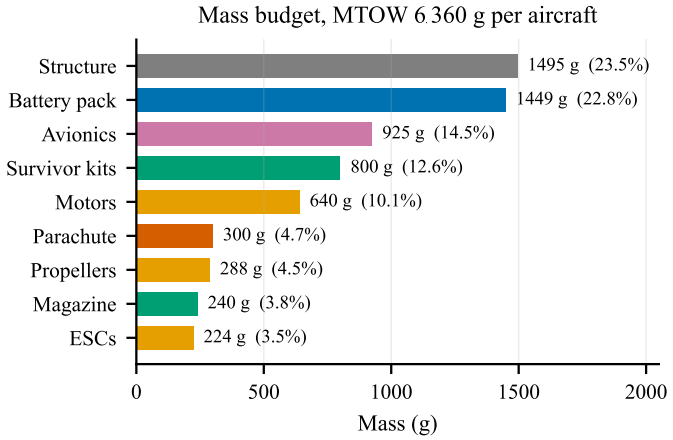


Fig. 1. Mass budget per aircraft, from the sizing model. Avionics includes the wiring harness, and the magazine includes the release mechanism. At 19 081 g the fleet leaves 4 919 g of growth allowance against our internal 24 kg target, so the build could come in 26 % overweight and still stay inside the 25 kg regulatory limit.

with a mean delivery error of 5.5 m. Because release conditions dominate, they had to compute the release point continuously against wind and aircraft velocity during flight [11]. A multirotor can instead slow almost to a stop before releasing, which removes the largest error term entirely. A fixed-wing aircraft cannot do this, and it is the main reason we chose a multirotor. Our delivery budget in Section IV-I is based on it.

E. Positioning

Real-time kinematic (RTK) correction from a local base station is standard practice, and it is the only way to get sub-metre relative positioning without an external network [12]. Since our system cannot use the internet or mobile data, it has to carry its own base station.

IV. Proposed System Architecture

A. Design point

Table I gives the design point of each aircraft. All of the values come from the sizing model described in Section V and are regenerated automatically rather than copied by hand.

One entry is not yet confirmed for the aircraft we are building. The motors we have chosen (Section VIII) do not come with a published thrust curve, so the thrust-to-weight ratio is a requirement until we measure it on the thrust stand. The other entries follow from the mission: mass, energy, altitude, speed and payload.

The battery rows describe the pack the model was originally sized with. We have since replaced the 18-cell Li-ion pack we planned to build ourselves with a ready-made 6S 16 Ah lithium-polymer pack. Section IV-B gives its figures and our reasons. We will re-run the model with the measured mass of the new pack before first flight and update this table.

The thrust-to-weight ratio of 2.0, and a hover endurance of 15.3 minutes for a 7.6-minute mission, are safety reserves rather than performance targets. The failure we most want to avoid is an aircraft that cannot get back. The energy reserve is set out in Section V, and it is the reserve, not the mission itself, that determines the size of the battery.

Fig. 1 shows the mass budget behind these figures. Structure and battery together make up 46 % of the take-off mass. This is why the risk register tracks airframe mass and battery chemistry closely, while the choice of avionics matters much less.

The items in the budget add up exactly to the maximum take-off mass, and the model checks this. The largest optional item is the recovery parachute and its mount at 300 g, 4.7 % of the aircraft. It is the only mass carried to protect the airframe rather than to do the mission. We have kept this allowance in the sizing even though the parachute itself is deferred from this build (Section VIII). The aircraft is therefore sized to fly with the parachute fitted, so adding it later uses up growth allowance without changing the design point. Until then the aircraft is 300 g lighter than the 6.36 kg quoted here, and every margin in this section is better by that amount. The four relief kits weigh 800 g in total, 12.6 % of take-off mass. The mission fixes them at exactly 200 g each, so the payload is an input to the sizing rather than something we can trade. This is also why we cannot cut cost on propulsion and batteries the way we can on avionics: cheaper parts of the same size exist, but a smaller aircraft would not carry the payload.

B. Energy storage and the reserve policy

Section V works through the battery requirement in full: the reserve that sets its size, the minimum rated energy, and the continuous current that rules out most candidate packs. The reserve includes a second sweep because finding nothing on the first pass is a normal result, not a failure. It includes time to hover because three aircraft cannot all land at once.

We chose a ready-made LiPo pack over building our own Li-ion pack. We first planned the Li-ion pack for its lower

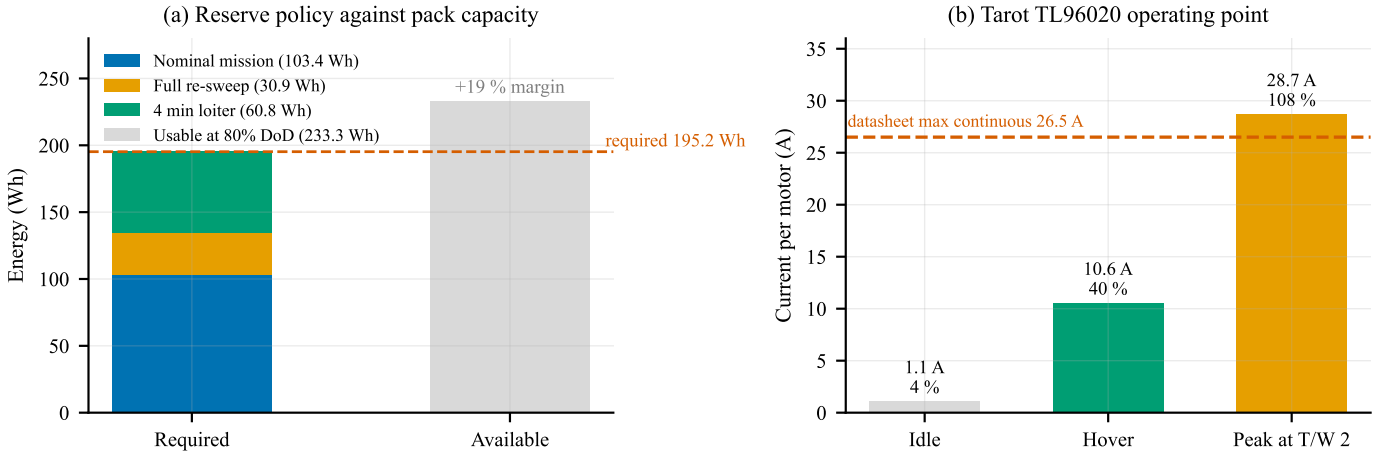


Fig. 2. (a) The energy reserve compared with what the pack can deliver: the mission itself, a second full sweep of the search area and four minutes of hovering, all within the 80 % usable capacity. (b) The selected motor compared with its datasheet. In hover it draws 40 % of its rated continuous current. At full design thrust it draws 108 %, which we accept only because full thrust is a short-lived reserve for gusts and manoeuvres, not a normal flight condition. This is the tightest margin in the propulsion system, and the thrust-stand test in P2 measures current as well as thrust to check it.

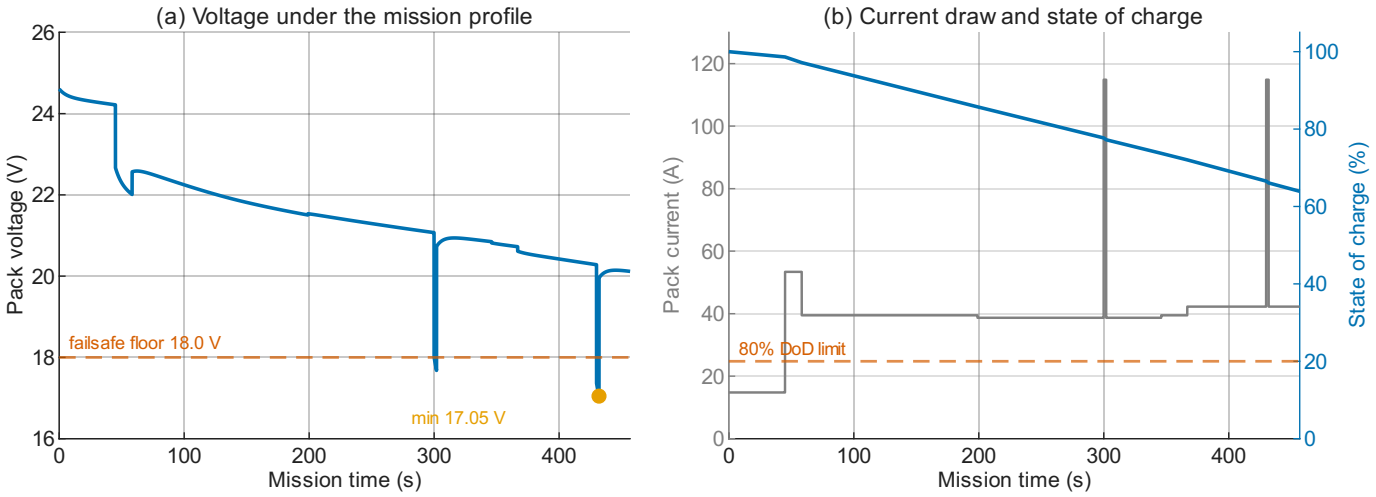


Fig. 3. Pack voltage over the mission, simulated with a second-order Thevenin cell model using the flight's own current trace plus two gust recoveries at full thrust. The pack drops to 17.05 V, below its 18.0 V failsafe threshold, while still two-thirds full and long before the 80 % discharge limit the reserve is based on. The dynamic sag is 1.2 V more than the static estimate because the slower diffusion effects add to the instantaneous resistive drop. A failsafe that looked only at instantaneous voltage would therefore send a healthy aircraft home in the middle of a search. The internal resistance here is assumed, not measured, and P2 replaces it with a measured value. This simulation was done for the earlier 6S3P Li-ion pack; we will repeat it for the LiPo using its measured sag before flight.

mass, longer cycle life and local availability, while noting that LiPo was electrically better. We changed our minds for three reasons. A suitable 6S LiPo pack is now sold by an Indian retailer, so sourcing no longer favours Li-ion. Buying a pack means we do not have to spot-weld eighteen cells in a student lab, which was the most hazardous step in the build. And the electrical advantage is large: a 25 C, 16 Ah pack can supply 400 A against a peak demand of about 136 A, whereas our own pack would have supplied about 110 A, and the LiPo has roughly half the voltage sag.

We could not simply swap one pack for the other, because a heavier aircraft needs more power to hover. We ran each candidate through the sizing model at its own mass, keeping the motors the same:

A 12 or 13 Ah pack fails both the endurance requirement and the landing-queue reserve, and about 14 Ah is the smallest that passes. The 16 Ah pack is the smallest suitable size on sale in India. The price of the extra mass is thrust-to-weight. The motors were sized for 2.0 at 6.36 kg, so at 6.8–7.1 kg they give only 1.8–1.9, unless the P2 thrust stand shows at least 3.55 kgf per motor. We will weigh the pack when it arrives and measure its voltage sag under load. Until then, Fig. 3 still describes the Li-ion pack. The battery failsafe has been set for the LiPo using standard values, and we will check it against the measured sag before any flight.

The reserve also has to be checked at the mass the aircraft actually flies. In the model the Li-ion pack is rounded up to a buildable 18 cells, which puts the take-off mass about 0.7 kg

TABLE II
FLIGHT PACK CANDIDATES, EVALUATED AT THEIR OWN MASS

Pack	kg	Wh	T/W	Endur.	Queue, Wh
Li-ion 6S3P (was)	1.45	292	2.00	2.02	+1.1
LiPo 12 Ah	1.42	266	2.01	1.85*	−3.7*
LiPo 13 Ah	1.52	289	1.98	1.96*	−0.5*
LiPo 14 Ah	1.64	311	1.94	2.06	+2.5
LiPo 16 Ah HV	**1.88–2.21**	**365**	**1.8–1.9**	**2.1–2.3**	**+5 to +9**

Endurance is hover time divided by mission time (must be at least 2.0). Queue is what remains of the 20 % reserve after the third aircraft waits 120 s and allows for two 53 s descents. * Fails. The maker does not publish the mass of the 16 Ah pack, so we give a range based on 165–194 Wh/kg.

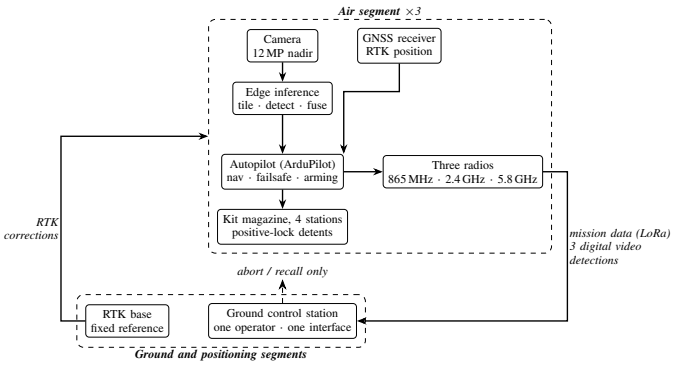


Fig. 4. System architecture. The only commands the ground station can send are abort and recall; it has no way to retask an aircraft, change a waypoint or order a drop. Detection runs on board, so the downlink exists for the operator, not for the detection software.

above the point where the sizing loop converges. We therefore re-evaluate the requirement at the final design point. There it is about a fifth higher, and the minimum rated energy rises with it. The pack still meets the requirement at the final mass, but with a noticeably smaller margin than a check at the converged point would show, which is why we run the check after the loop rather than inside it.

C. Segment architecture

The system comprises three segments, shown in Fig. 4.

1) Air segment: The air segment is three identical quadrotors. Each has a Holybro Kakute H7 autopilot running ArduPilot [14], mounted on a 4-in-1 speed controller whose current sensor feeds the autopilot's battery monitor. Each also has a dual-band RTK receiver based on the u-blox ZED-F9P, which tracks GPS L1/L2, GLONASS G1/G2, Galileo E1/E5b and BeiDou B1/B2 [15]. The receiver, its helical antenna and a compass come as one 20 g unit. The autopilot has no compass of its own, so heading comes from the one in the receiver. The rest of each aircraft is a downward-facing 12 MP camera, a Raspberry Pi 5 with a 26 TOPS AI accelerator, a magazine holding four kits with mechanical locks, a 5.8 GHz digital video radio, an 865 MHz LoRa radio, and a 2.4 GHz ExpressLRS link that gives the safety pilot control and an abort path independent of the autonomy software.

2) Ground segment: A single ground control station gives the operator one interface for all three aircraft. The station cannot retask an aircraft, change its waypoints or order a drop. Abort and recall are the only commands it can send, and they are a deliberate exception. This is built into the software rather than enforced by a setting: the code for other commands is not present in the build we fly, so it cannot be enabled by mistake.

3) Positioning segment: We use our own RTK base station. During setup it takes its position from its first three-dimensional fix and holds that as its reference. The aircraft may take off once they have a three-dimensional fix, but they do not report a survivor position until they have an RTK fixed solution. Detections made before then are tagged with the less accurate float solution and recalculated once the fix is obtained.

Setting the base this way saves time at a cost. The base is a second ZED-F9P with a survey-grade antenna on a tripod. A survey-in would locate it to within a few centimetres, but it takes many minutes, and skipping it removes that time from setup. The cost is that a base set from its first fix carries that fix's absolute error, which is around a metre, and every survivor coordinate we report is shifted by the same amount. RTK still gives the relative accuracy in Table V: each aircraft knows its position relative to the base to within centimetres, wherever the base thinks it is.

For this mission we think the trade is worth it. A rescue crew will navigate to the reported coordinate with its own receiver, which is only accurate to about a metre anyway, so a shared offset of under a metre makes little practical difference. Twenty minutes of survey-in, on the other hand, would come straight out of the flying time. When there is time to set up the base before the flying window opens, we will run the survey. We will always do so in P7 and P8, because a measured accuracy figure has to be checked against something more accurate than the system being measured.

The base sends corrections over its own 865 MHz LoRa radio, and each aircraft receives them on a matching radio. On the aircraft this radio connects to the companion computer, because all of the autopilot's serial ports are already in use. The companion forwards the RTCM 3.x correction stream to the autopilot as MAVLink, and ArduPilot passes it on to the receiver using `GPS_INJECT_TO` [14]. Corrections therefore travel on a different path from both the command link and the video link. We limit the base to MSM4 messages at 1 Hz so that the stream fits within the radio's capacity. If corrections are lost, the receiver falls back to SBAS, so the degraded mode needs no extra hardware.

D. Geolocation error budget

The accuracy of survivor coordinates is the main output of the system, so the error budget is the most important analysis in the design. We take the total error as the root sum of squares of the GNSS position error, the remaining camera alignment (boresight) and lever-arm error, attitude error, terrain-model error and timing error.

1) How much the receiver matters: One might expect positioning quality to dominate, which would make RTK essential. The budget shows otherwise. In case C there is effectively

TABLE III
GEOLOCATION ERROR BY RECEIVER CLASS, 200 000 SAMPLES

Receiver	RSS	CEP50	CEP95
RTK fixed, 1 cm	0.88 m	**0.73 m**	1.53 m
SBAS, 0.60 m	1.07 m	0.89 m	1.85 m
Standalone multi-band, 1.0 m	1.33 m	1.11 m	2.31 m

no GNSS position term, because at 1 cm it is negligible next to everything else. Two terms that no receiver can improve, unmodelled error at 0.70 m and the size and centre of the target at 0.50 m, account for 95 % of the variance between them (Fig. 6a).

Putting a position term back in for less accurate receivers therefore changes the result only a little. A Monte Carlo simulation over the full budget, treating each term as an independent two-axis Gaussian, gives:

RTK improves CEP50 by 0.16 m over SBAS. Every receiver class meets the 5 m delivery requirement with more than three times margin. The simulation is checked against the analytic result for a circular Gaussian, CEP50 $= 0.8326\times$RSS, and stops with an error if the two disagree. The RTK row also reproduces, independently, the 0.75 m requirement set by attitude error.

There is a further point. Because the base station takes its position from its own first fix rather than a survey-in, its absolute position is off by about a metre, and every coordinate we report carries that offset. RTK keeps the relative accuracy in Table V, with each aircraft placed to within centimetres of the base, but against true position the difference between RTK on an unsurveyed base and a good SBAS receiver is even smaller.

We are keeping RTK anyway, for a different reason. The budget does not fail without it, but centimetre-level relative positioning fixes the geometry between survivors, aircraft and landing pad precisely, and that is what we need in P7 and P8 to measure detection recall and delivery error against surveyed ground. In other words, RTK is needed to produce reliable test results, not to meet the mission accuracy. If funding did not stretch to it, dropping to SBAS would cost 0.16 m of CEP50 and, more importantly, our ability to measure the system accurately.

All three aircraft carry RTK. We had first planned to fit it to one aircraft only, since all three are identical and one would be enough to characterise the design. However, a ZED-F9P rover with antenna and compass now costs about the same as the ordinary receiver plus separate compass it replaces, so fitting RTK to every aircraft costs no more than fitting it to one. It also means any aircraft can serve as the test aircraft, and losing one in testing does not leave us without RTK.

2) Required receiver accuracy: The comparison above covers three receivers. Reviewers have often asked a broader question: why not plain GPS, or NavIC, which is Indian and costs about a fifth as much as the RTK unit? We answer it with numbers below.

TABLE IV
RECEIVER CLASSES AGAINST THE 5 M DELIVERY REQUIREMENT, 200 000 SAMPLES

Class	σ_g	CEP50	CEP95	
RTK fixed	0.01 m	0.74 m	**1.53 m**	clears
SBAS / GAGAN	0.60 m	0.89 m	**1.85 m**	clears
Multi-band autonomous	1.00 m	1.11 m	**2.31 m**	clears
GPS L1 only	2.50 m	2.21 m	**4.59 m**	marginal
NavIC SPS	5.00 m	4.23 m	**8.79 m**	*fails*

NavIC is not an alternative to RTK, because the two work at different levels. NavIC is a satellite constellation, like GPS, GLONASS, Galileo and BeiDou, whereas RTK is a correction technique applied to whatever constellations the receiver tracks. The useful comparison is between classes of receiver performance, and NavIC appears in it at its standalone accuracy, since that is how we would use it.

Everything in the budget apart from the receiver adds up to 0.88 m (Fig. 6a). This is a floor that no receiver can lower. Adding a receiver error σ_g in quadrature and converting to the 95 % radius gives the condition directly:

$$\mathrm{CEP95} = 1.7308\sqrt{\sigma_{\mathrm{rest}}^2 + \sigma_g^2} \;\le\; 5\,\mathrm{m} \implies \sigma_g \le 2.75\,\mathrm{m}. \tag{1}$$

So any receiver with horizontal error below 2.75 m meets the delivery requirement, and any receiver worse than that does not, whatever else we change.

Three conclusions follow. First, NavIC at its published standalone accuracy does not meet the requirement, so it cannot replace the receiver on any aircraft. It is still useful as an extra constellation tracked alongside the others, which improves satellite geometry at no cost if the module already supports it. Second, GPS alone is marginal at 4.59 m against a 5 m requirement: it passes on paper but leaves nothing in hand for a day with poor satellite geometry. Third, SBAS meets the requirement at 1.85 m, which makes it an acceptable fallback when RTK corrections are lost.

The SBAS figure deserves a closer look. We assumed 0.60 m, which is optimistic; field results for this kind of augmentation are closer to 1.0–1.5 m. Using Equation (1), the requirement is still met at either value, with CEP95 of 2.31 m and 3.01 m respectively, so the conclusion does not depend on the assumption. It would start to matter at around 3 m, which is why we measure the actual horizontal accuracy on the built aircraft in P5 instead of relying on the datasheet.

A second receiver on each aircraft would provide moving-baseline heading. That is a separate question, and we have deferred it (Section VIII).

Combining several frames costs us almost nothing. At the 40 m survey altitude the camera's footprint is 31.4 m long in the direction of flight, so at 8 m/s a target stays in view for 3.9 s. At a conservative 2 Hz detection rate that gives about eight separate views of each target per pass, and at the 3 Hz rate we assume for confirmation it gives twelve. We use these

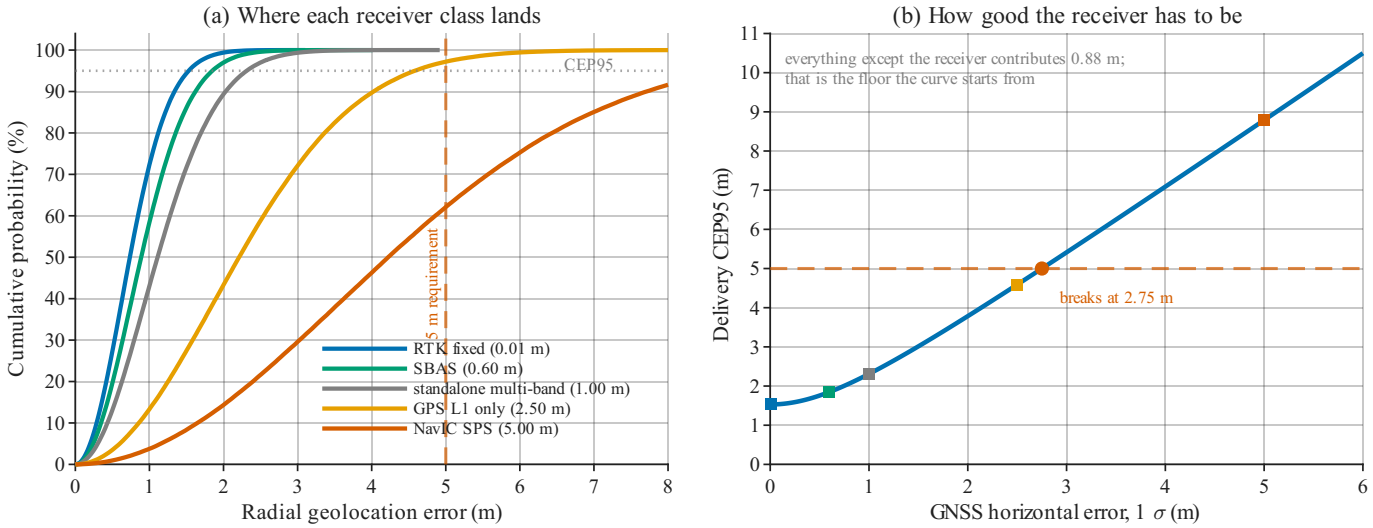


Fig. 5. (a) Radial error distribution for five receiver classes over the full geolocation budget, 200 000 Monte Carlo samples each. (b) The same result as a criterion: delivery CEP95 against the receiver's own horizontal error, with the 5 m requirement and the 2.75 m limit from (1). The markers are the five classes from panel (a), in the same colours. The curve does not start at zero because 0.88 m of the budget comes from terms no receiver can reduce.

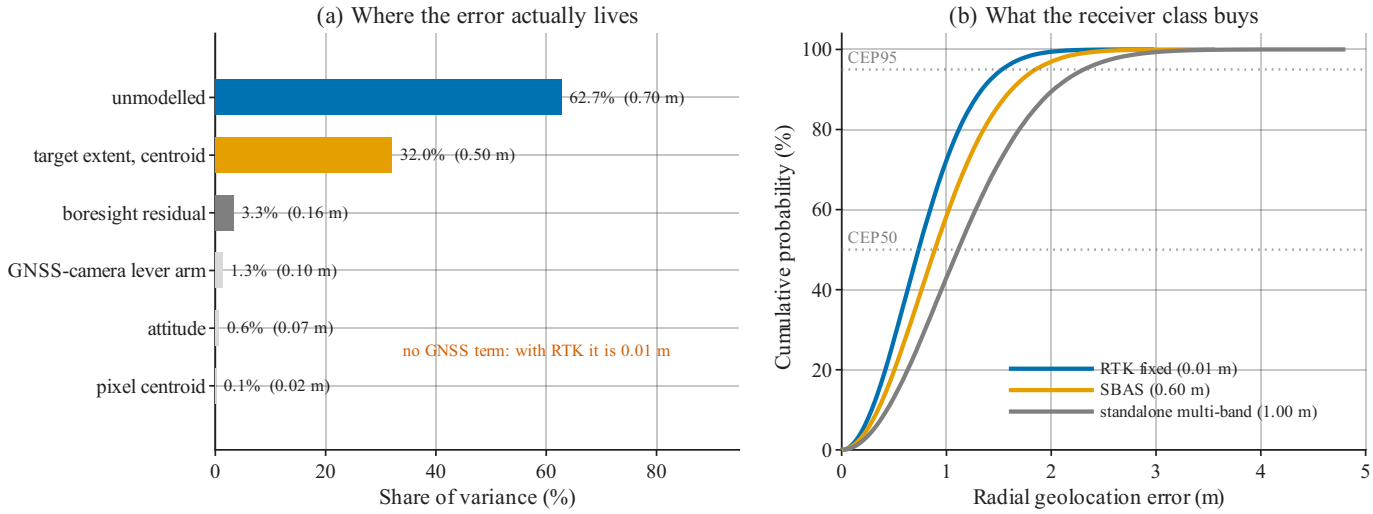


Fig. 6. (a) Sources of geolocation error. Unmodelled error and the target centre account for 95 % of the variance. There is effectively no GNSS position term, since with RTK it is 1 cm. (b) Radial error distribution by receiver class, from 200 000 Monte Carlo samples over the full budget. RTK and SBAS differ by only 0.16 m of CEP50, because the larger terms are ones neither receiver can reduce.

views both to confirm a detection, which reduces false alarms, and to average its position.

Fig. 7 shows both effects. Panel (a) shows how much positioning quality matters when everything else is single-frame: losing the RTK fix increases the error almost tenfold, and a plain 3-D fix at 3.67 m RSS uses up most of the 5 m delivery allowance before release and hover errors are even counted. Panel (b) shows that combining frames helps, but with diminishing returns. It divides the random terms by $\sqrt{n}$ but does nothing for alignment error or target size, so the curve leaves the ideal $1/\sqrt{n}$ line and flattens out by about five frames. Twenty frames are not twice as good as five, and the eight to twelve views we get anyway are already past that point.

E. Perception and inference budget

The camera is a 12.3 MP module, 4056×3040 pixels at a 1.55 µm pitch, behind a 6 mm lens. We tile each frame at full resolution into 48 crops of 640×640 with 20 % overlap, without shrinking it first. At 2 Hz this is 96 detector runs per second. Published measurements for this accelerator with the same model and input size are 130–160 per second, so it fits with some margin.

We considered halving the resolution before tiling, which would cut the number of crops from 48 to 12. That option made sense for a slower accelerator with about a quarter of the throughput, but it throws away half of the linear resolution the camera provides. The number of tiles depends only on the

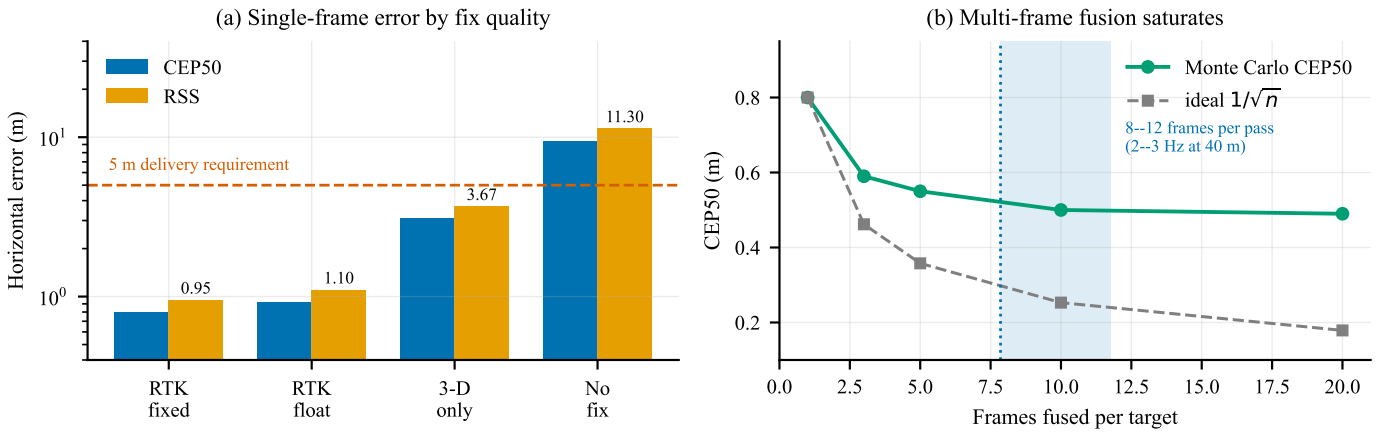


Fig. 7. Geolocation error, from a Monte Carlo run through the same projection code the aircraft flies. (a) Single-frame horizontal error against GNSS fix quality, typical attitude, on a logarithmic scale; the dashed line is the 5 m delivery requirement. (b) Effect of fusing multiple observations of the same target, RTK fixed.

TABLE V
GEOLOCATION ERROR BY CONFIGURATION

Case	Configuration	RSS error
A	No RTK, single-frame	3.09 m
B	RTK, single-frame	∼1.3 m
C	RTK + multi-frame fusion + calibrated ground plane	**0.91 m**

Case C assumes moving-baseline heading. That receiver is deferred (Section VIII). Using compass-based attitude instead gives 0.94 m, only 0.03 m worse, which is why we think the deferral is affordable.

number of pixels, so it does not change with altitude. What changes with altitude is the size of the target in the image, and the effect is larger than it first appears:

AGL	Tiling	Target (px)	Area (px^2)	COCO class
40 m	**native**	**39**	**1 498**	medium
60 m	native	26	666	**small**
40 m	2× downsample	19	375	**small**
60 m	2× downsample	13	166	**small**

The target in this table is a person in water, showing only head and shoulders about 0.4 m across. This is the hardest case, so we design for it. An adult lying on a rooftop appears about four times larger.

What matters is which size class the target falls into, more than the exact pixel count. Detection results are normally reported by size class, and accuracy on small objects is usually much lower than on medium ones for the same model. Only the top row is above the COCO small-object threshold of 32^2 px^2. That is why we fly at 40 m and tile at full resolution, and why we need both changes together: either one on its own leaves the target in the size range where detectors perform worst.

Two modelling choices lie behind these figures. We computed the optics for the actual 1.55 µm sensor we are buying, not for a generic sensor with the same number of pixels, because it is the pixel size that sets ground resolution. And we took the target as a person in water rather than lying on dry ground, which is four times smaller in the image and the cautious choice for a flood mission.

Finer ground resolution narrows the camera's footprint, so covering the area takes more passes. At 40 m the extra passes add only seconds to a mission with minutes to spare, and the mission is actually shorter than at 60 m because the aircraft has less height to climb.

This throughput figure is the one most likely to change once we test real hardware, so we have scheduled and budgeted benchmarking on the purchased parts before the design is frozen.

F. Division of the computing hardware

Each aircraft carries a camera, a single-board computer and an AI accelerator. On an aircraft this small it is fair to ask why these are not one part. The reason is that each does a job the others cannot.

The accelerator is an M.2 module that connects only over PCIe. It has no camera input, no image processor and no general-purpose cores; it simply runs a compiled network on data that the host computer passes to it. The camera produces a raw Bayer stream, which has to be demosaiced, white-balanced and scaled before it is a usable image. The tiling our detection approach depends on, cutting each frame into 48 overlapping crops, sending them to the accelerator and mapping the detections back onto the full frame, also has to run on the host. So does everything else the aircraft does: the mission state machine, MAVLink routing, the data link, claiming and releasing survivors, the delivery manoeuvre and the telemetry sent to the ground station.

The three parts are therefore a camera, a computer that processes images and runs the mission, and an accelerator that runs the network. Removing any one of them would break the detection chain rather than simplify the aircraft.

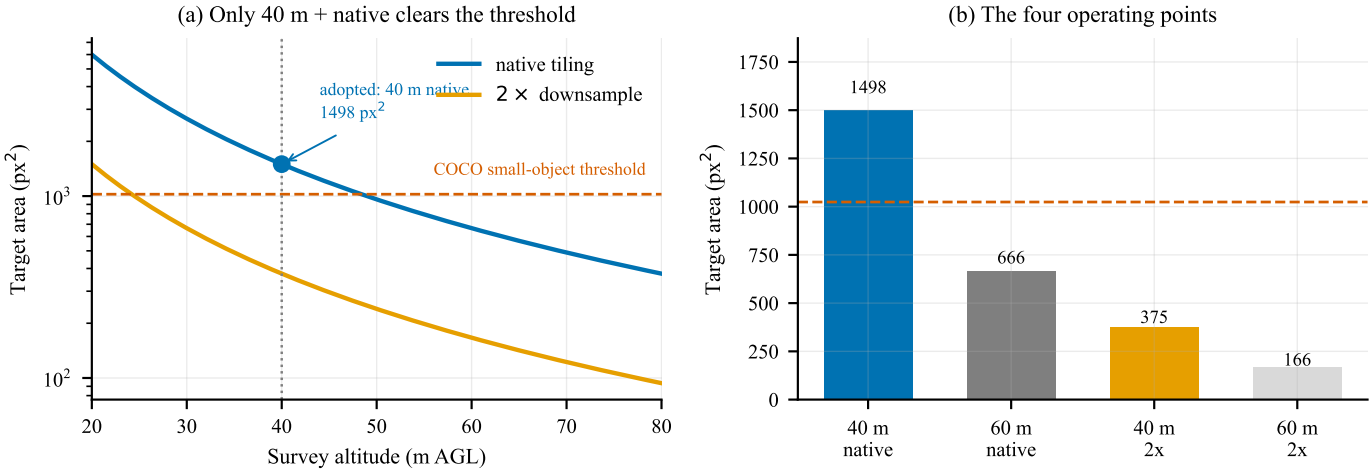


Fig. 8. Target area against survey altitude for a person in water showing head and shoulders. (a) Only our chosen setting, 40 m with full-resolution tiling, is above the COCO small-object threshold; halving the resolution puts the target below it at every altitude. (b) The four candidate settings. Either change on its own leaves the target in the size range where published detector accuracy drops sharply.

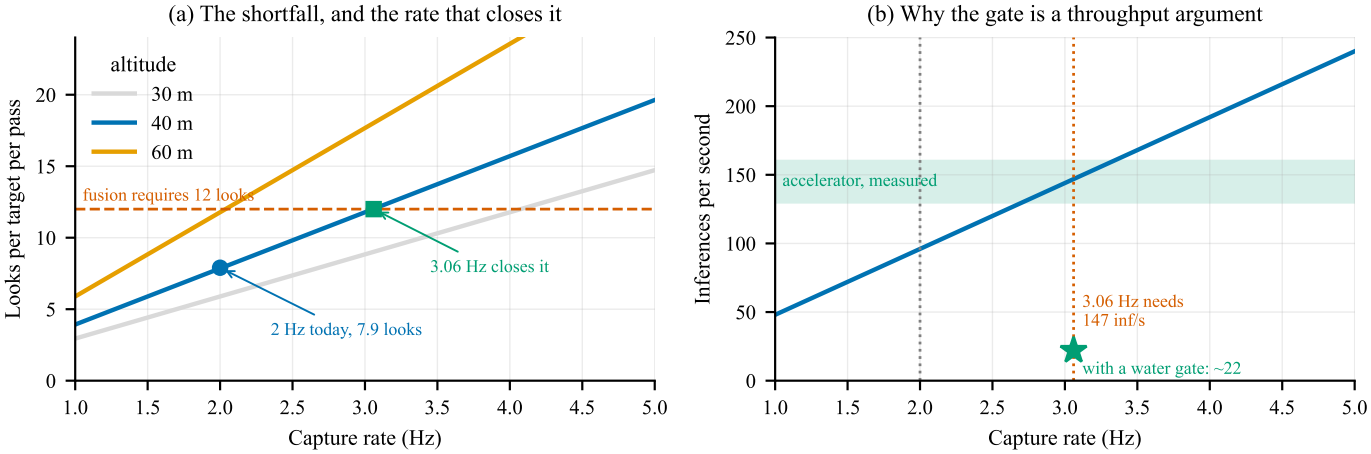


Fig. 9. (a) Views of each target per pass against capture rate. Combining frames needs twelve views; our 2 Hz rate gives 7.9, and 3.06 Hz would give twelve. Flying lower makes the shortfall worse, because the footprint shrinks faster than the pass gets shorter. (b) The cost of the higher rate. With 48 tiles, 3.06 Hz means 147 detector runs per second, against 130–160 measured for the accelerator, which is at or beyond its limit. A quick first check that discards tiles of plain open water would free up that capacity, so the case for a two-stage detector is about throughput rather than accuracy.

There is one alternative that would remove a part, and we considered it carefully. Some camera sensors now include an AI engine on the chip. They run the network themselves and output detections instead of images, use about 40 % less power than a separate accelerator, and would remove the accelerator altogether. They do not suit this mission. These on-sensor engines hold a model of only a few megabytes and run once per frame at a fixed input size. Our design needs **48** inferences per frame, because a survivor covers only a few tens of pixels in a 12 MP frame and tiling is what makes such small targets detectable (Section IV-G). A sensor that cannot tile its own image cannot do this, and we would still need the host computer for everything listed above.

G. Onboard processing pipeline

Fig. 10 shows what runs where. The key point is that the autopilot flies the aircraft and the companion computer does not interfere with it during the sweep. The search lines are uploaded during setup as an ordinary AUTO mission, and the aircraft would fly the whole search pattern even with the companion computer switched off. The companion watches, detects, and takes control only to make a delivery. This is a short GUIDED manoeuvre, after which the aircraft returns to the same AUTO mission.

Keeping this boundary has two benefits. If the companion crashes or overheats, we lose detections but not control of the aircraft. And the search path can be reproduced from the mission file alone, which lets us check the coverage and separation results in Section VI-B directly.

Detection never holds up the flight. The detector works on captured frames while the aircraft carries on along its line. If frames are lost because the computer overheats or the model runs slowly, we lose some views of a target, and the twelve-view requirement leaves room to absorb some of that. The

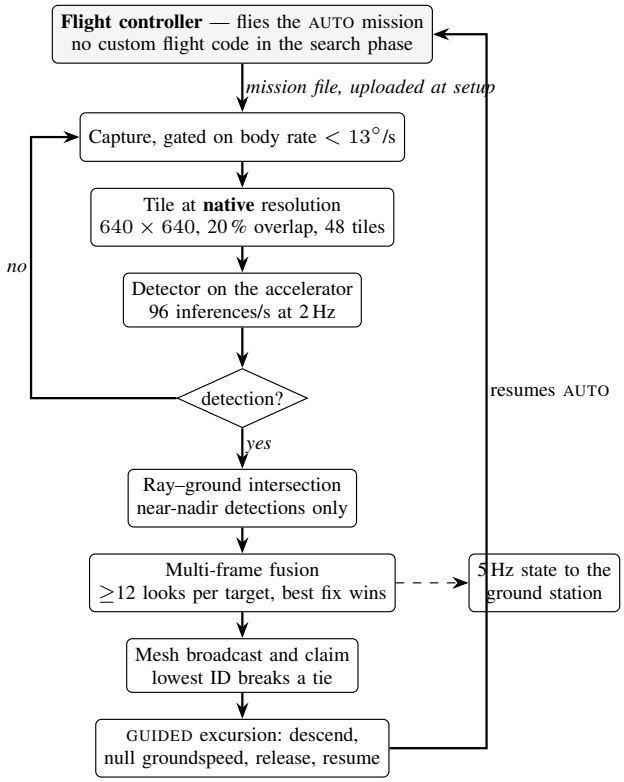


Fig. 10. Onboard processing. The autopilot flies the search pattern on its own; the companion computer detects, locates and claims survivors, and interrupts the mission only to make a delivery. Detection runs alongside the sweep and never holds it up, so a missed frame costs one view of a target, not a waypoint. Frames are not captured during turns, where the flat-ground assumption is weakest and motion blur is worst. One pixel covers 13.63 mdeg on this sensor, so at a $1/1000$ s exposure a rotation rate of $13.6°$/s gives one pixel of blur; we set the limit at $13°$/s to stay below it.

sweep does not stop and the mission does not get longer.

The throughput figures above are published measurements, not our own. We have not yet benchmarked the accelerator and host together with our model and our tiling, so the inference budget is still unconfirmed. Benchmarking on the purchased parts in P5 is the check before we freeze the design, and the risk register (Table X) treats the figure as open until then.

H. Communications

The competition rules require the ground station to show a live camera feed from each aircraft, and the system must work without any external network. We use three radio links, each in a different band, and size each one so that it has margin to spare at the edge of the flying area.

Video (digital, 5.8 GHz, one channel per aircraft). Each companion computer draws its detections onto the camera image and sends it using wfb-ng [16] on an RTL8812EU radio. The operator and the judges therefore see what the aircraft's software sees, rather than a raw feed. Each of the three aircraft has its own channel. At the ground station there is one receiver per channel, each with a 14 dBi panel antenna pointed at its aircraft and an omnidirectional antenna on its second port. The second port also handles the short uplink the protocol needs, so neither port is left unconnected.

TABLE VI
LINK BUDGET AT THE 600 M GEOFENCE

Link	Received	Margin
Digital video, 5.8 GHz	−56.3 dBm	29.7 dB
ExpressLRS, 2.4 GHz	−71.6 dBm	36.4 dB
LoRa, 865 MHz, SF7	−68.7 dBm	**54.3 dB**
802.11 data link, 5.8 GHz	−73.3 dBm	*8.7 dB*

The companion computer has no hardware video encoder, so it has to encode the video in software. We have accepted this load and limited it: each computer encodes only its own stream, the detector runs on the accelerator rather than the processor, and if the processor gets busy it is the video resolution and frame rate that drop, never the number of detection tiles. In P5 we will measure how much processor capacity is left with both running.

Mission data and coordination (SX1262, 865–867 MHz). This link carries position, flight mode, battery, fix quality, survivor tags and delivery status as one compact binary message per aircraft per second, instead of raw MAVLink. The band is licence-free in India for short-range devices. At SF7 it has a 54 dB margin, the largest of any link, which suits the link that carries survivor coordinates.

That margin also lets us choose a faster mode. Working out the channel load from data size and bit rate alone gives the wrong answer, because for short packets LoRa's preamble and header take most of the airtime: at SF7 the fleet's messages fill a quarter of the channel, not the few per cent the bit rate suggests. The same radio can run GFSK at 50 kbps, which uses 19 dB of the spare margin, and that is the mode we use, with SF7 kept as a fallback at a lower message rate. Section V-I works through both, and Table VIII gives the results.

Command and abort (ExpressLRS, 2.4 GHz). This is the only link that can command the aircraft, and it connects straight to the flight controller rather than going through the autonomy software. If the companion computer hangs or is compromised, the safety pilot still has control. This follows from how the hardware is wired, not from any software permission.

ExpressLRS has supported MAVLink natively since version 3.5, so the receiver that gives the safety pilot manual control also carries the autopilot's command and telemetry stream. On the ArduPilot side it needs only a serial port set to MAVLink v2 [17]. RTK corrections do not use this link; they come from our base station on 865 MHz (Section IV-D).

We originally planned an 802.11 mesh network, but dropped it because of its link budget. At the edge of the geofence it had only 8.7 dB of margin, the weakest link in the system, and it would have carried video, telemetry and coordination all at once. A mesh does not fix that, because a relay hop is just as weak. Rule 8.17 forbids relying on an external network for coordination between drones, but it does not require the

aircraft to talk to each other directly, and coordination through our own ground station counts as a locally deployed system under rule 8.5. The cost of this arrangement is that one aircraft cannot relay for another that has lost its link, and the link-loss failsafe already covers that case. The benefit is that no band carries more than one job.

I. Payload delivery

Kits are released 6 m above the ground, and only once the aircraft's remaining ground speed is below 0.3 m/s. Under these conditions the fall itself adds about 0.32 m to the delivery error. Combined with geolocation and hover error, the total is about 1.5 m (root sum of squares) with RTK working. The release height is low enough to keep wind drift under a metre in realistic conditions, and high enough to stay safely clear of people and debris.

Each kit station has a mechanical lock, so a power failure cannot release a kit. We treat this as safety-critical and will not cut cost on it.

J. Mission sequence

Fig. 11 shows the autonomous mission sequence. Three features are worth pointing out. First, the take-off delays are written into the mission file as a pause before each take-off command, rather than relying on the operator's timing. This is what increases the closest approach at launch from 1.31 m to 64.80 m (Fig. 13). Second, every abnormal condition ends in the same way: link loss, low battery and operator abort all lead to return-to-launch. Third, the battery response has two stages. At the reserve threshold the aircraft returns to launch, and at a second, lower threshold it lands immediately where it is. If a battery cannot get the aircraft back, it is better to land under control as soon as that becomes clear than to come down uncontrolled on the way home.

V. Sizing and Analysis

This section shows how we arrived at the design point, not just the result. Every figure is regenerated from `tools/sizing-model/rescueswarm_sizing_model.py`, and the section itself is generated from it, so any change to a constant in the model appears here too.

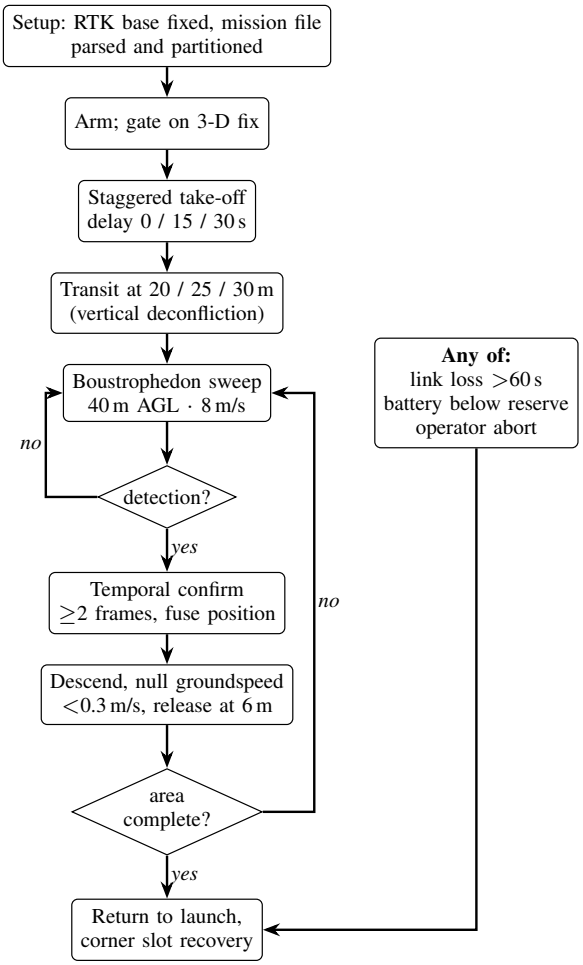


Fig. 11. Autonomous mission sequence. The take-off delays are set in the mission file rather than left to the operator, and all three abnormal conditions lead to the same return-to-launch.

A. Constants and where they come from

Symbol	Value	Basis
ρ	1.150 kg/m^3	∼30 °C at 300 m density altitude; an Indian summer field, not sea level
FM	0.60	Rotor figure of merit, mid-range of the 0.5–0.7 literature band
η_{mot}	0.82	BLDC efficiency near the hover operating point
η_{esc}	0.95	Electronic speed controller
P_{avio}	55 W	Companion computer, autopilot, GNSS, camera, radios, servos
DoD	80%	Usable depth of discharge; land at 20 % state of charge
T/W	2.0	Static thrust-to-weight, a reserve policy

We chose two of these values cautiously. Air density is taken

for a hot day at altitude rather than at sea level, which increases the power required, and the figure of merit is taken from the middle of the published range rather than the top. We would rather find out now that the design is tight than discover it on the flying field.

B. Hover power, from momentum theory

The rotor disk area for 4 rotors of diameter 18 in ($D = 0.4572$ m) is

$$A = N\pi\left(\frac{D}{2}\right)^2 = 0.6567 \text{ m}^2,$$
$$\text{disk loading} = \frac{m}{A} = 9.69 \text{ kg/m}^2.$$

Momentum theory gives the induced power needed to produce thrust T over that disk area [18]. Dividing by the figure of merit FM, the ratio of ideal induced power to actual shaft power, accounts for blade drag and uneven inflow and gives the shaft power:

$$P_{\text{shaft}} = \frac{T^{3/2}}{\text{FM}\sqrt{2\rho A}} = \frac{(62.4)^{3/2}}{0.60\sqrt{2(1.150)(0.6567)}} = 668 \text{ W},$$

where $T = mg = 62.4$ N at the 6.36 kg maximum take-off mass. Adding the drivetrain and the avionics load,

$$P_{\text{elec}} = \frac{P_{\text{shaft}}}{\eta_{\text{mot}}\eta_{\text{esc}}} + P_{\text{avio}} = \frac{668}{0.779} + 55 = 913 \text{ W},$$

which at the 21.6 V pack nominal is a hover current of $I = P/V = 42.2$ A. Power loading is 7.0 g/W.

C. Peak power and the current the pack must supply

Induced power scales as $T^{3/2}$, so at the design thrust-to-weight of 2.0 the shaft power rises by $2.0^{3/2}$:

$$P_{\text{peak}} = \frac{(2.0\,T)^{3/2}}{\text{FM}\sqrt{2\rho A}\,\eta} + P_{\text{avio}} = 2481 \text{ W} \implies I_{\text{peak}} = 114.8 \text{ A}.$$

It is this current, more than the energy, that rules out most candidate packs. It has to be supplied continuously rather than in short bursts, because the aircraft must be able to hold 2.0 times its weight, not just reach it briefly. The modelled pack supplies 135 A, a margin of 18 %, and each motor's share sets the speed controller rating at 29 A peak.

This is also why a smaller pack does not work. At 13.5 Ah the peak is 8.51 C and hover is 3.13 C; a pack of half the capacity would need twice the C-rate to deliver the same current. A larger capacity keeps the discharge rate down.

D. Propulsion

$$T_{\text{total}} = (T/W)\,mg = 125 \text{ N} = 12.72 \text{ kgf},$$
$$T_{\text{motor}} = 3.18 \text{ kgf}.$$

Hover needs 1.59 kgf per motor, which is 50 % of maximum thrust. This is inside the 45–55 % range where both propeller efficiency and thermal margin are reasonable. Well below that range, the propulsion system would be larger and heavier than necessary; well above it, there would be too little thrust left to hold attitude in gusts.

With the motor we have chosen, hover is comfortable but peak thrust is not. The Tarot TL96020 is rated for 26.5 A continuous. In hover each motor draws 10.6 A, which is 40 % of that rating. At the design thrust-to-weight it draws 28.7 A, or 108 %, which is above the continuous rating. We accept this only because full thrust is a short-term reserve, not a flight condition: the aircraft hovers at about half throttle and reaches this current for a few seconds at most, when recovering from a gust or manoeuvring hard. It is still the tightest margin in the propulsion system, so the P2 thrust stand measures current as well as thrust to show whether the motor reaches 3.18 kgf within its rating. The datasheet also gives a motor mass of 168 g, compared with the 160 g the sizing model assumes, a difference of +32 g per aircraft that we will confirm when we weigh the parts in P5.

E. The coupled mass and energy solve

Pack mass, take-off mass and mission energy all depend on one another: a heavier pack raises hover power, which raises the energy needed, which raises the pack mass again. The model repeats the calculation until the values settle.

The reserve is what fixes the result. The pack must provide enough usable energy for the mission, one complete second sweep of the search area, and four minutes of hovering:

$$\underbrace{103.4}_{\text{nominal}} + \underbrace{30.9}_{\text{re-sweep}} + \underbrace{60.8}_{\text{loiter}} = 195.2 \text{ Wh usable},$$

against 233.3 Wh available at 80% depth of discharge, a margin of 19 %. Whatever its chemistry, a candidate pack must therefore be rated for at least **244 Wh** at **6S** and supply **114.8 A continuously**.

The model was sized with 18 cells in 6S3P: 292 Wh, 13.5 Ah, 1449 g, 21.6 V nominal (18–25.2 V). Hover endurance is 15.3 min against a 7.6 min design mission consuming 103.4 Wh. The pack now specified is a bought 6S 16 Ah LiPo (Section IV-B); these figures are re-run on its measured mass before first flight.

At 6.36 kg per aircraft the fleet weighs 19.08 kg, against the 25.0 kg regulatory limit, a margin of 24 %.

F. Camera and optics

All of the optics follow from three numbers: pixel count, pixel size and focal length. The active sensor area is the pixel count times the pixel size,

$$w = 4056 \times 1.55\,\mu\text{m} = 6.287 \text{ mm},$$
$$h = 4.712 \text{ mm}, \qquad d = 7.86 \text{ mm},$$

and the fields of view are calculated exactly, without the small-angle approximation:

$$\theta = 2\arctan\left(\frac{s}{2f}\right)$$
$$\Rightarrow \text{ HFOV} = 55.3^\circ,\ \text{VFOV} = 42.9^\circ,\ \text{DFOV} = 66.4^\circ.$$

By similar triangles through the lens, the ground sample distance at altitude H is

$$\text{GSD} = \frac{p\,H}{f} = \frac{1.55\,\mu\text{m} \times 40\,\text{m}}{6\,\text{mm}} = 1.03 \text{ cm/px},$$

giving a 41.9×31.4 m footprint. One pixel subtends 14.80 mdeg.

We consider two target sizes and design for the smaller one. An adult 1.7 m tall lying on a rooftop or dry ground covers 165 px, or 7,960 px^2. A person in water shows only head and shoulders, about 0.4 m across, and covers 39 px, or 1,498 px^2, roughly a quarter of the area. We design for the person in water, because this is a flood system and that is the posture we expect. The larger figure is the easy case and is given only for comparison. Section IV-E bases the tiling decision on the water figure.

Focus is not a problem. The hyperfocal distance,

$$H_{\text{hyp}} = \frac{f^2}{Nc} + f = 4.15 \text{ m},$$

taking the circle of confusion as two pixels, puts everything from 2.08 m to infinity in acceptable focus. The aircraft never flies below 30 m, so a fixed lens set once is in focus at every altitude we use. A focusing mechanism would only add a moving part, extra power draw and a failure we could not detect in flight.

Motion blur is not a limit either. Keeping blur under one pixel requires $t_{\text{exp}} \leq \text{GSD}/v$, which at 8 m/s is 1.29 ms, a shutter speed of $1/774$ s. We already require $1/1000$ s or faster.

Rolling shutter has a small effect. Reading out the frame takes 25 ms, during which the aircraft moves, giving 0.20 m of skew from top to bottom, about 19 px. This does not matter for detection, but it does matter for geolocation: unless the software corrects for the row in which a detection appears, its position will be biased along the direction of flight.

Number of views is where the budget falls short. A target stays in the frame for the footprint length divided by ground speed, so at 40 m and 2 Hz it is seen 7.9 times, against the 12 views we want for combining frames. Getting that many needs

$$r \geq \frac{n\,v}{D} = 3.06 \text{ Hz},$$

The options are a faster capture rate, a lower ground speed, a higher altitude, or accepting fewer views. Raising the capture rate is the cheapest. Flying lower shrinks the footprint faster than it shortens the pass, so it actually increases the rate needed; flying higher makes the ground resolution coarser; and flying slower makes every pass longer.

The cost of a faster capture rate is processing, not energy. At 48 tiles per frame it needs 147 detector runs per second, against the 130–160 per second measured for the accelerator, which is at or beyond its limit. A two-stage approach could recover this. A first stage that only asks whether a tile is plain open water is judging texture, not looking for a person, so it still works on a shrunken image even though, as Section IV-E shows, a person would be lost at that size. At 96×96 input it handles about 44 times fewer pixels than the detector, and discarding the open-water tiles would leave the equivalent of about 22 full detector runs. This first stage would run on the accelerator we already carry rather than on an extra processor, because the saving is in detector runs, not power: computing uses less than one per cent of the aircraft's hover power.

G. Payload release ballistics

A kit of 200 g tumbling with drag coefficient 1.05 and reference area 117 cm^2 has ballistic coefficient $\beta = m/(C_d A) = 16.3$ kg/m^2 and terminal velocity

$$v_{\text{term}} = \sqrt{\frac{2mg}{\rho C_d A}} = 16.2 \text{ m/s}.$$

Integrating the fall with quadratic drag from a 6 m release gives 1.15 s to impact at 9.7 m/s, with a crosswind drift of 3.24 m at 3 m/s and 6.13 m at 6 m/s.

The speed of the aircraft at release matters far more than the height. A remaining ground speed of 0.5 m/s moves the impact point by about 0.53 m, while a whole metre of height error moves it by less than 4 cm. This is why we hover before dropping: a multirotor can stop before release, and a fixed-wing aircraft cannot.

H. Structure, geometry and control authority

Tip clearance of 30 mm between adjacent 18 in rotors sets a diagonal wheelbase of 689 mm and an arm length of 0.345 m. At the design thrust-to-weight the root bending moment is

$$\mathcal{M} = T_{\text{motor}}\, L = 10.7 \text{ N m}.$$

For a 25×23 mm carbon tube, $I = \pi(D_o^4 - D_i^4)/64 = 5438\,\text{mm}^4$, so the bending stress is $\sigma = \mathcal{M}c/I = 25$ MPa against roughly 600 MPa for unidirectional carbon, a safety factor of about 24.

Bending is therefore not what limits the structure, and we should not design as if it were. The joints and clamps, and vibration fatigue where the arm meets the frame, matter much more. A tube that passes the bending case by a factor of 24 is being chosen for stiffness and ease of handling, not strength.

Dropping one kit from a magazine that sits off the centreline moves the centre of gravity by less than 2 mm, which takes only a few tens of grams of extra thrust to correct. The amount is tiny, but it happens suddenly and disturbs the attitude control. We avoid this by mounting the magazine on the centre of gravity and holding position briefly after each release.

I. Radio links

Three kinds of traffic have to be carried, and they have very different needs. Video needs a high data rate but can be lost occasionally. Command and abort need little data but must not be lost. Mission data and coordination also need little data and must keep working even when the other two fail. We first planned to carry all three on one 5.8 GHz 802.11 mesh network, but dropped the idea after working out its link budget.

1) Link margin: Free-space path loss at the 600 m design slant range follows Friis,

$$\text{FSPL} = 20\log_{10}\!\left(\frac{d}{1\,\text{km}}\right) + 20\log_{10}\!\left(\frac{f}{1\,\text{MHz}}\right) + 32.44, \quad (2)$$

but path loss alone does not tell us much, since it is always higher at 5.8 GHz than at 865 MHz. What decides whether a link works is the margin left over,

$$M = P_{\text{tx}} + G_{\text{tx}} + G_{\text{rx}} - \text{FSPL}(f, d) - S_{\text{rx}}, \quad (3)$$

TABLE VII
ADOPTED RADIO LINKS, EVALUATED AT THE 600 M GEOFENCE

Link	P_{tx} (dBm)	ΣG (dBi)	S_{rx} (dBm)	FSPL (dB)	M (dB)
Digital video, 5.8 GHz	27.0	20	-86	103.3	**29.7**
ExpressLRS, 2.4 GHz	20.0	4	-108	95.6	**36.4**
LoRa, 865 MHz SF7	14.0	4	-123	86.7	**54.3**

where S_{rx} is the receiver sensitivity for the modulation we actually use, not the best figure on the datasheet. Table VII gives (3) at the geofence for our three links.

2) Why we dropped the single mesh: Applying (3) to the 802.11 setup we had planned, with 20 dBm into 5 dBi antennas at each end and an MCS0 sensitivity of -82 dBm, the same 103.3 dB of path loss leaves only 8.7 dB of margin. That would have been the weakest link in the system and also the one carrying the most data, so the link most likely to fail would also be the one whose failure costs most. With three separate radios, every link has more than 29.7 dB, an improvement of 21.0 dB on the video link alone (Fig. 12a).

A mesh does not solve this, even though that was the original reason for choosing one. Relaying through a second aircraft replaces one 600 m link with two shorter ones and reduces path loss. But each hop uses the same radio, antennas and sensitivity, so the relay is only as reliable as the aircraft carrying it, and the things that break the direct link, such as a banked airframe, a person between the antennas or rain, tend to affect both hops. A mesh improves average throughput but not the worst case, and the worst case was our problem.

The 802.11 link would not be useless. It only reaches its sensitivity limit at 1.64 km and would work well in good conditions (Fig. 12b). But rain, people in the way, a banked airframe and misaligned antennas all eat into the margin, none of them is included in (3), and 8.7 dB is not enough to cover them. Our video link has 29.7 dB at the same distance and only runs out at 18.4 km.

3) Sizing the coordination channel: Splitting the links has a consequence that the margin table does not show. The mesh would have offered about 235 kbps of non-video capacity per aircraft, whereas LoRa at SF7 on a 125 kHz channel carries 5469 bps for the whole fleet, 129 times less. We had not compared the two designs against the same traffic, so at first we missed this.

What matters is time on air, not bit rate. For the short packets this mission sends, LoRa's preamble and header take up most of the time. The number of symbols in a packet of PL bytes is

$$n_{\mathrm{sym}} = n_{\mathrm{pre}} + 4.25 + 8 + \left\lceil \frac{8PL - 4\mathrm{SF} + 44}{4\,\mathrm{SF}} \right\rceil (\mathrm{CR}+4), \quad (4)$$

each lasting $2^{\mathrm{SF}}/\mathrm{BW}$. A 36-byte message therefore takes 77.1 ms on air, compared with the 52.7 ms its data alone would suggest, so the overhead is about a third of the total.

Our first plan sent aircraft state, task coordination and detection reports as three separate messages, each at its own rate. That meant three preambles and three headers per aircraft per second, and it used 28.4 % of the channel, more than the 25 % limit we set to leave room for retries and a fourth node. Combining them into one 36-byte message per aircraft per second, with a detection added when there is one, brings this down to 25.3 %. That still leaves almost no spare capacity: the maximum message rate is 1.0 Hz against the 1 Hz we need.

TABLE VIII
COORDINATION CHANNEL, THREE AIRCRAFT, ONE 36 B FRAME EACH AT 1 HZ

Mode	Rate	Frame	Occupancy	Margin
GFSK 50 kbps	50 kbps	7.8 ms	**2.3 %**	35.3 dB
LoRa SF7, 125 kHz	5.5 kbps	84.2 ms	25.3 %	54.3 dB

The answer is already in Table VII. The 865 MHz link has 54.3 dB of margin at the geofence, far more than this mission needs; the spreading factor is buying range we do not use. The SX1262 can also run GFSK with the same chip, antennas and 14 dBm output. At 50 kbps its sensitivity is -104 dBm instead of -123 dBm, which gives up 19 dB of the spare margin for a 9.1 times higher data rate. Each message then takes 7.8 ms, channel use drops to 2.3 %, and the margin is still 35.3 dB, more than the video link has, while using only a tenth of the channel.

We therefore use GFSK normally and LoRa SF7 as a fallback, switching on the same radio without any hardware change. The fallback is genuinely slower: at 25.3 % channel use it supports only 1.0 Hz, so when we switch to it we lower the message rate.

4) Order of degradation: We have arranged the links so that they fail in a sensible order, following their margins. Video is allowed to fail first. Because it is digital, the picture stays clean until the margin runs out and then frames start to drop, and losing it costs the operator a view but does not affect the aircraft's decisions. Coordination degrades next, from GFSK to LoRa SF7 and then to a lower message rate. Command and abort come last, on ExpressLRS with 36.4 dB of margin, and are never traded away: if that link is lost, the aircraft carries out its failsafe rather than waiting for the link to return. All the bands are licence-free in India: the 865–867 MHz short-range device band for coordination, and the 2.4 and 5.8 GHz ISM bands for command and video, each used well within its power limit.

There are two limitations. All these margins assume free space, so they are best cases: reflections off water, a low ground antenna clipping the Fresnel zone, and the airframe blocking the signal in a turn all happen in practice and none is modelled. And the GFSK sensitivity comes from the datasheet, not a measurement. In P5 we will measure packet loss against distance on the built aircraft, and only then will Table VIII be backed by evidence.

J. Limits of this section

Every figure in this section is modelled. The thrust figures rely on a motor with no published thrust curve, and the

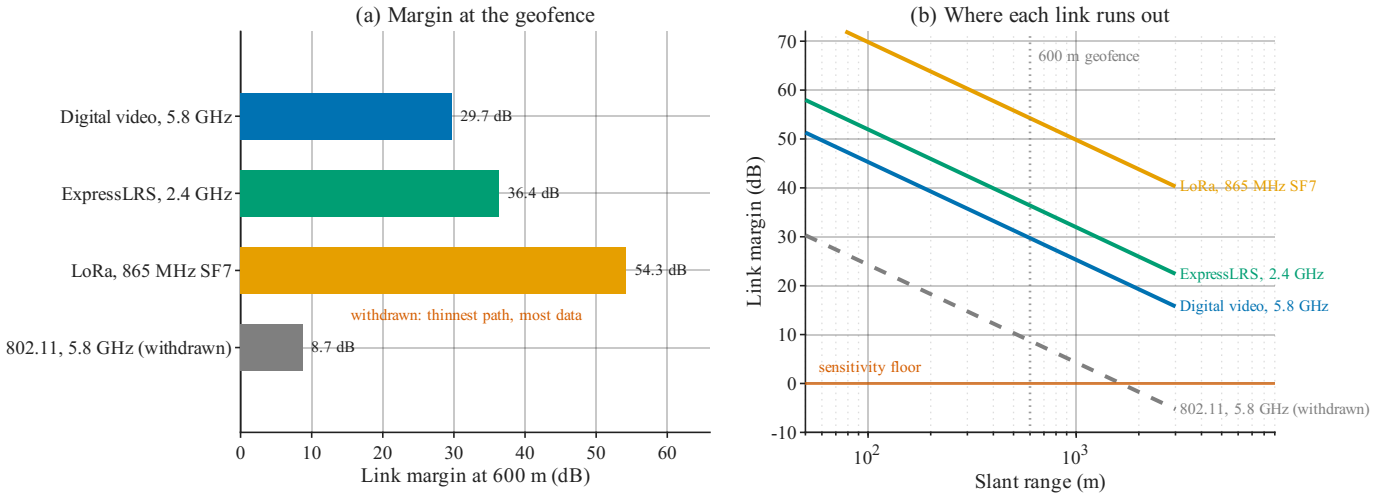


Fig. 12. (a) Link margin at the 600 m geofence, from (3). The 802.11 mesh we dropped is included for comparison. (b) Margin against distance. Each of our links is used at a small fraction of its maximum range; the dropped mesh would not have been.

detection figures assume a recall that has not been measured. The sizing is consistent and its arithmetic can be checked, but that does not show that the aircraft can actually do these things, and we do not claim that it can.

VI. Methodology and Work Plan

The work is divided into ten phases. P1–P5 cover analysis, purchasing and integration, and P6–P10 are flight tests of increasing difficulty. The eight-week flight-test window covered by P6–P10 is the tightest constraint on the schedule, and the plan is arranged to protect it.

A. Onboard health monitoring

Two more small models run on board, separate from survivor detection. Both work on simple sensor signals and are small enough to run on a microcontroller, so they take nothing from the accelerator, which Section IV-E shows is already busy.

The first detects propulsion problems. We already log current for each motor and vibration of the airframe for the P6 tests. The same signals can reveal an unbalanced propeller, a worn bearing or a loose mount before it causes a failure in flight. That matters when we are flying twelve motors on three aircraft over an eight-week campaign, and the training data comes from flight testing anyway.

The second estimates the health of the battery. The reserve in Section V uses a fixed rule: never discharge below 20 %. That rule suits the chemistry but ignores cell ageing and temperature, so it is too cautious when the pack is new and not cautious enough when it is old. A model that uses voltage, current and temperature under load can estimate the usable energy left directly. Since the reserve is what determines the pack size (Section IV-B), a better estimate is the cheapest way to gain margin or flying time.

For both models, a wrong answer costs some energy or an unnecessary precautionary landing, but never a missed survivor. We only allow small models where that is true, which is why neither is part of survivor detection.

TABLE IX
Programme phases

Ph.	Title	Principal exit criterion
P1	Requirements and sizing	Design point frozen; all models regenerate
P2	Ground segment	Single-operator interface demonstrated against three simulated aircraft
P3	Autonomy	Deterministic plan generation; deconfliction verified in simulation
P4	Fault handling	Link-loss and low-battery behaviour verified by fault injection
P5	Procurement and build	Three airframes assembled and weighed; thrust stand complete
P6	First flight	Hover, trim, vibration and thermal characterisation
P7	Perception field trials	Detection recall measured on real imagery against fabricated targets on RTK-surveyed ground
P8	Delivery trials	Delivery CEP measured against a surveyed target mat
P9	Full mission rehearsal	End-to-end mission with all three aircraft, no operator intervention
P10	Setup and contingency	Setup drill timed over repeated runs; margin recovered

B. Verification strategy

Our approach to verification grew out of a kind of mistake we kept making. Four times during the preparatory work, we found a configuration file that was correct, tested and reviewed, but was never actually used by the running system. In one case a checked set of failsafe parameters was never loaded by any

simulation, so every test had run on the default settings, in which the low-battery action was switched off. In another, a message-routing configuration started without errors and then forwarded no messages at all for weeks. In every case the file had passed review and its tests.

Our response is a working rule rather than a tool: we do not count anything as demonstrated until the real configuration has been run end to end and the resulting values have been read back from the running system. Every claim in this proposal is labelled with the kind of evidence behind it:

- *Measured*: observed on a running system, in software or in hardware-in-the-loop simulation.
- *Modelled*: calculated by a model whose inputs are assumptions, and regenerated automatically on every change.
- *Assumed*: neither measured nor modelled, and listed as a risk.

An automated check regenerates every model output we publish and compares it with the committed version, so changing a model without updating its output causes the check to fail.

VII. Preliminary Results

We have done a good deal of work already. We summarise it below and say what kind of evidence supports each result, so that readers can check it.

A. Measured in simulation

- *Low-battery failsafe.* With three software-in-the-loop autopilots running at once, the failsafe returned each aircraft to the pad. Return-to-launch triggered at 10 809 mAh against a planned 10 800 mAh.
- *Three aircraft on one ground station.* A single telemetry router delivered three separate system IDs to one ground station port, and all three arrived.
- *Separation at launch.* Putting take-off delays in the mission file, instead of relying on the operator's timing, increased the closest approach between airborne aircraft during launch from 1.31 m to 64.80 m (Fig. 13). Both figures are recalculated directly from the recorded telemetry.
- *Separation during the sweep.* Away from the pad the aircraft never came closer than 29.19 m. Different transit altitudes of 20, 25 and 30 m keep them apart vertically regardless of the horizontal plan.
- *Separation at recovery.* Placing the landing slots at the corners of the pad instead of in a row increased slot spacing from 1.22 m to 2.61 m at no cost. The measured closest approach rose from 0.83 m, where the aircraft overlapped, to 2.27 m, leaving 1.22 m between airframes (Fig. 14). Separation while the aircraft wait above the pad improves in the same way.
- *Ground station.* All eight required operator displays worked with three autopilots running at once, using offline map tiles (Fig. 16). We tested the video wall with three streams in a separate run with the media server switched on. It was off when the screenshot in Fig. 16 was taken, which is why the video panels show as offline.

1) No margin at the pad edge: In Fig. 14 the corner slots sit exactly on the inner square. The nominal clearance to the edge is 0.523 m, one airframe radius, so a stationary aircraft on its slot lies entirely inside the pad and meets the rule.

However, this leaves no margin at all. The rules do not allow any part of an aircraft outside the pad during launch and landing. A touchdown error of ± 0.5 m, the figure we use elsewhere for landing accuracy, could put up to half a metre of an airframe over the edge.

The two requirements conflict: moving the slots inward to gain edge margin reduces the spacing between aircraft, which is the reason we moved them to the corners in the first place. To settle this we need to measure the actual touchdown error, which we will do in P6. We mention it here because the layout looks safe in the figure but is only safe on average.

B. Modelled

Endurance, hover power, mass budget, thermal margin, link budget, delivery error and geolocation error are all modelled, and all regenerate from source. The geolocation Monte Carlo agrees with a separate analytic calculation to within one per cent. We have also checked the code that converts a detection into a ground position against a second, independent formulation, and the two agree to 7.8×10^{-10} m.

C. Obstacles

The aircraft fly their sweeps without commands from the ground, so avoiding buildings has to be built into the design. Three measures do this, and there is one thing the aircraft cannot do.

Flying altitude clears low buildings only. The sweep is flown at 40 m above ground, which is well above the two- to four-storey buildings typical of rural and semi-urban flood plains. It is not high enough for cities. Chennai and Mumbai, where serious flooding happens, have many buildings taller than 40 m.

The aircraft also cannot simply fly higher. The survey altitude is set by detection: at 40 m a person in water covers 1498 px^2, above the COCO small-object threshold, but at 50 m the same person covers only 959 px^2 and falls below it. No altitude both clears a 60 m building and lets this camera resolve a 0.4 m target. Any obstacle taller than the sweep height therefore has to be avoided by going around it.

Avoiding obstacles sideways relies on marking them in advance. The operator draws the search area on site, and ArduPilot's exclusion zones let a pylon, a tower or a tall building be marked as an area the aircraft will not enter. The autopilot enforces these zones itself, so they still apply even if the mission plan is wrong. But they only work if someone has seen the obstacle and drawn it.

Height above the surface is measured directly. A downward-facing lidar gives the true height above whatever is below the aircraft. This matters most at release. The kit is meant to drop from 6 m, but a barometer measures height above the launch

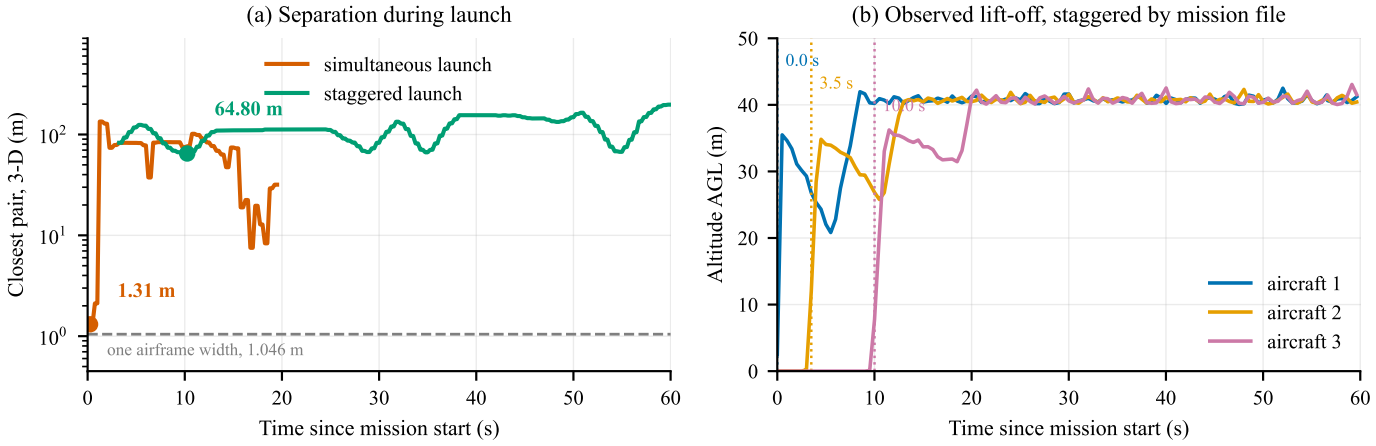


Fig. 13. Separation at launch, recalculated from the recorded telemetry of three ArduCopter software-in-the-loop autopilots. (a) Distance between the closest pair of airborne aircraft in three dimensions, on a logarithmic scale. Launching all three at once brings two aircraft within 1.31 m, less than one airframe width (marked), while staggered take-off keeps the closest pair 64.80 m apart. (b) Lift-off times observed in the same recording. The mission file commands delays of 0, 15 and 30 s; in this recording the aircraft left the pad at 0.0, 3.5 and 10.0 s, and the figure shows the observed times.

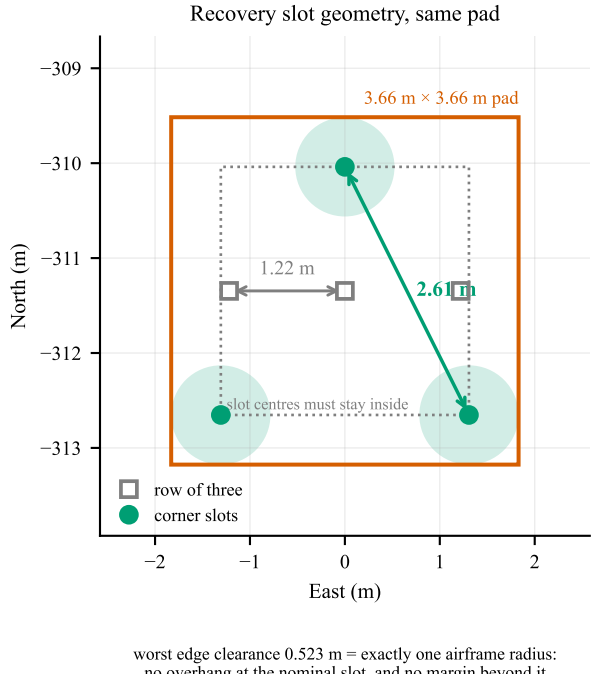


Fig. 14. Landing slot layout, drawn from the two recorded flights. Slot centres must stay half an airframe inside the pad edge, which confines them to a 2.61 m square. A row across that square gives 1.22 m spacing; its corners give the full 2.61 m. The shaded discs show the 1.046 m airframe footprint at the corner slots. The measured closest approach rises from 0.83 m, where the airframes overlap, to 2.27 m, on the same pad and at no extra cost.

point, which over floodwater of unknown and changing depth is not the same thing. Section IV-I shows that impact speed and landing error both grow with release height, so a kit dropped from what the aircraft thinks is 6 m but is really 15 m behaves quite differently from our analysis. The lidar reaches about 7 m outdoors, so it picks up the surface during the descent and controls the release. It does not look forward and is not an obstacle sensor.

The aircraft cannot detect obstacles ahead of it. Nothing on board can see a structure in its path, so a tall building that has not been marked as an exclusion zone is invisible to the system, and the aircraft would fly into it.

This limits where the system can be used. As designed, it is for open water and low-rise flooding, where 40 m is above everything and an exclusion zone can cover the occasional pylon. It is not suitable for flooded cities with tall buildings. Adding a forward sensor would not be enough on its own: the aircraft could only stop or turn, not climb, so operating among tall buildings would need a route planner that finds a way around obstacles at a fixed altitude, which is a separate project. A 360-degree lidar costs around INR 10,000 and weighs 190 g, well within the 4.9 kg growth allowance, and could be useful as a last-resort stop. However, an unfunded and untested sensor flown for the first time at the competition would add more risk than it removes. If we buy one, we will test it in P6.

D. Assumed or not yet implemented

Several figures are still assumptions: the remaining camera alignment error after calibration, the RTK accuracy we will actually achieve in the field, and the battery's internal resistance in Indian summer temperatures. Each will be measured in P5–P8. Detection recall on real flood imagery is also an assumption, and because the field campaign is deferred (Section VIII) we do not currently have a plan to measure it. That is why it is at the top of the risk register.

One ground station function is not implemented, and Fig. 16 shows this. The abort and recall buttons record the operator's command in the mission log but send nothing, because the command link from the ground is not yet wired. For now, recovery depends on the safety pilot's transmitter. We will close this in P4 using the 2.4 GHz ExpressLRS link, since the separate sub-gigahertz radio is deferred. That makes the

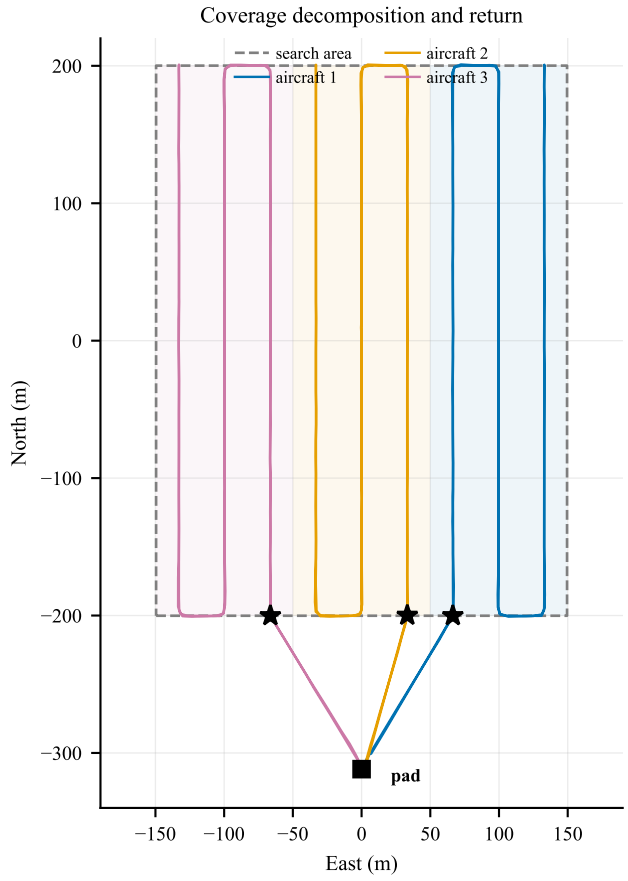


Fig. 15. Area division and return paths, drawn from the recorded telemetry. Each aircraft is given an equal-area strip (shaded) and sweeps it back and forth. Each strip is covered twice in opposite directions so that camera alignment error cancels out instead of adding up. Stars mark where each sweep ends. By trying all four combinations of starting side and direction and choosing the one that finishes closest to home, all three aircraft end within 117–130 m of the pad. Under our earlier rule, two of the three ended 516 m and 540 m away. The path length is the same, and the saving comes when the batteries are lowest.

ExpressLRS link essential for safety, not just a convenience for manual control.

E. Evidence status

Nothing in this project has flown yet. Every result we describe as measured was measured in simulation or in software. That rules out many logic and integration errors, but none of the physical ones. The main purpose of the build is to turn the modelled and assumed figures in this section into measured ones.

VIII. Scope of This Build

This document describes the technical system. The funding request is a separate document. This section covers what the build includes and leaves out, because several arguments earlier in the paper depend on it.

A. Deferred items

We have deferred the following items, not dropped them as unnecessary. We list them so that reviewers can see what the lowest-cost version gives up, and because two of them limit what this paper can claim.

The field data campaign and the ground-truth equipment are both deferred. As a result, detection recall in this paper is modelled, not measured, and we currently have no funded way to measure it. Of everything in this list, this has the biggest effect on what the system can be shown to do, and it is the first item we would restore if funding allowed.

The second GNSS receiver on each aircraft is deferred. It would provide moving-baseline heading, but Section IV-D shows that using compass-based attitude instead costs only 18 mm in the case-C budget. The real reason to want it is protection against the compass being disturbed by high-current wiring nearby, which the budget does not model.

Our receiver does make this possible later: two ZED-F9P receivers on one aircraft, one in moving-base mode, provide heading, and ArduPilot supports this setup. We have deferred it for mass and cost, not availability, so heading currently comes from the compass in the rover unit. In P6 we will measure the heading accuracy on the built aircraft, and if it is worse than about 1° we will fit the second receiver.

The separate sub-gigahertz safety radio is deferred. This depends on the main link being configured correctly: it must run in the mode that carries both manual control and MAVLink telemetry over one link and one autopilot serial port. In the alternative transparent-serial mode the link carries only telemetry, and the aircraft has no manual control. We will confirm the correct mode on the bench in P2; if it does not work, the safety radio is no longer optional.

Recovery parachutes are allowed by the rules but not required. A crash costs 50 points against 10 for landing outside the zone, so deferring parachutes means risking about forty points on the chance of a crash.

Spares are limited to parts that break in crashes. We keep spare propellers, motors, arms, plate stock and battery packs, but no spare autopilot, GNSS receiver or camera. We accept the risk that a failure of one of these during the flight-test window would cost us time, and our only protection is ordering replacements quickly.

Flight insurance is deferred, but whether it is required must be checked against the rulebook and the Drone Rules before any flight. Unlike the other items, this is not just a performance trade-off.

B. A limitation of our ground truth

We will make the ground-truth targets ourselves rather than buy them. The survivor targets are clothed forms of the right size and shape; a detector responds to outline, scale and contrast against the background, not to anatomy. For two items, making our own is actually better. We can make dummy kits weigh exactly the 200 g that rule C6 requires, and donated clothing gives a wider range of colours and contrasts than

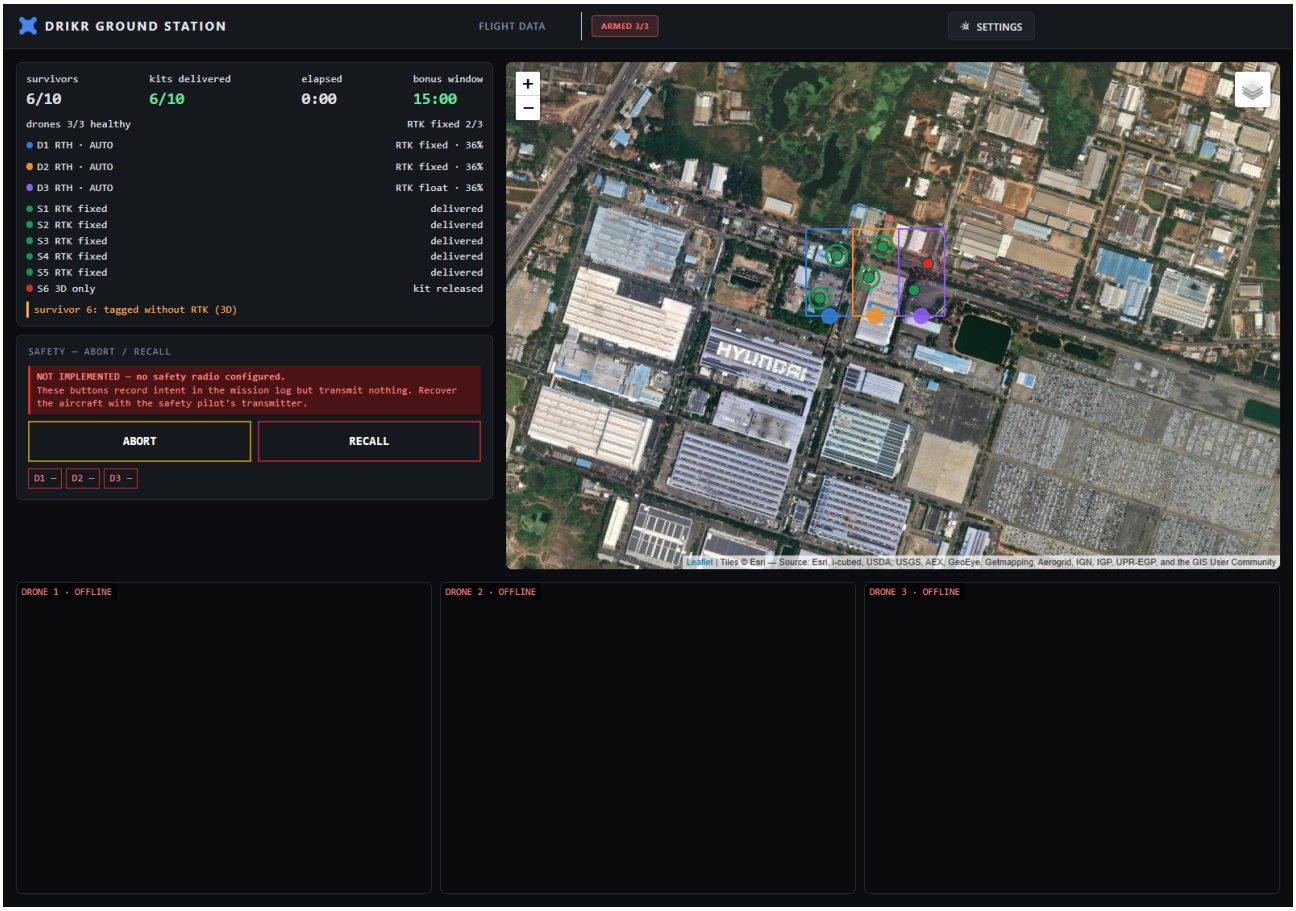


Fig. 16. Ground control station, an unedited screenshot of the running software. The telemetry is simulated: a test driver generates MAVLink and 5 Hz mission status for three aircraft, so that every display can be tested before any aircraft exists. No real aircraft, autopilot or radio is involved. The screen shows the three search areas on a satellite map served from a local cache, so no network is needed; three aircraft armed and in AUTO, two with an RTK fixed solution and one in float, a degraded case the display must show clearly; and the mission timers, where 15:00 is the threshold for early completion rather than the flight-time limit. The screenshot was taken after delivery, with six survivors found and six kits dropped, so the per-survivor list is visible. Five survivors are tagged RTK FIXED and the sixth 3D ONLY. The display shows this difference instead of presenting all six positions as equally reliable, because a rescue team going to the sixth location will need to search a wider area (Section IV-D2). We have left the safety panel uncropped. It states that abort and recall are not yet implemented: they are recorded in the mission log but nothing is transmitted, and recovery is by the safety pilot's transmitter, because the command link from the ground is not yet wired (Section VII-D). The video panels show as offline because the media server was not running when the screenshot was taken.

matching bought sets, which helps test how robust detection is.

We will survey the calibration markers with our own RTK rover. This is normal practice and good enough for the purpose, but it means the reference positions we measure geolocation against are themselves RTK-derived. The errors in the two are therefore correlated, and any geolocation accuracy we report from P7 and P8 will be slightly better than a comparison against an independently surveyed reference would show. We will say so whenever we report those figures.

IX. Security and Data Protection

Three features of the design also serve security, and we built them in from the start.

The ground station cannot command the aircraft. In the build we fly, the server has no command routes at all, so the operator interface cannot send a waypoint, a mode change or a release. This is stronger than access control, because there is nothing to bypass.

Command authority is kept apart from the autonomy software. ExpressLRS connects straight to the flight controller, not through the companion computer. If that computer hangs, is compromised or runs a model that misbehaves, the safety pilot still has control. This follows from how the hardware is connected.

Nothing depends on an outside network. All three links (Section IV-H) use licence-free bands with no infrastructure between the aircraft and the ground station, so there is no route for a remote attack.

A. Link security

The mission data link carries survivor coordinates, so it is encrypted. The video link uses wfb-ng's key-pair encryption, and ExpressLRS uses a binding phrase that is kept out of the code repository and is different for each aircraft. Neither adds

noticeable delay. An unencrypted link would also be harder to defend under the rules, since rule 8.4 prohibits public networks and an open link looks more like one.

In a disaster area the realistic threat to positioning is interference rather than deliberate spoofing: crowded spectrum, other responders' equipment, and signals reflecting off water and buildings. Tracking four constellations on two frequency bands gives some protection if one is degraded, and the operator checks fix quality before launch. We have not yet evaluated integrity monitoring on the receiver.

B. Survivor data as personal data

The system records the precise locations of identifiable people in distress, along with aerial images of them. Under the Digital Personal Data Protection Act 2023 this is personal data, and the people concerned are in no position to give or refuse consent when it is collected.

We therefore follow four rules about what happens to this data afterwards:

1) Coordinates are sent to the operator and kept only for the duration of the operation. They are deleted after handover to the rescue authority.
2) Images are kept only when they are needed as evidence for a specific measurement, and they stay on the aircraft rather than being transmitted.
3) We will not publish any image of an identifiable person, in this paper, in our results or in any dataset, unless the face and body are permanently obscured.
4) We train detection models on public datasets whose subjects agreed to that use, not on images of real casualties.

The fourth rule has a cost. It is part of the reason detection recall remains modelled rather than measured, and why our perception testing (Section VI-B) uses home-made targets and public sea imagery instead of real flood imagery, which would be more representative. We accept lower-quality measurement rather than collect data we should not hold.

X. Risk Register

The setup-time risk deserves particular attention. Our only evidence for it is a model with a modest margin, and if setup runs over, the whole mission is at risk. For this reason, timed bench drills are the first hardware activity in the plan.

XI. Expected Outcomes and Deliverables

1) Three flight-tested aircraft with a documented bill of materials, every part priced from a current listing.
2) A ground control station for a single operator, demonstrated end to end with real aircraft and no network.
3) A measured detection recall figure, using home-made survivor targets on surveyed ground, to replace the current assumption. The reference positions will be RTK-derived rather than independently surveyed, and we will say so when reporting the figure.
4) A measured delivery accuracy (circular error probable) against a surveyed target, to replace the current model.
5) Measured setup-to-launch times over repeated drills.
6) An open and reproducible record of results, in which every published figure can be regenerated from its model automatically.

XII. Future Work

This proposal covers one prototype build. The work could go further, and we outline the next steps here to make the limits of the present scope clear.

A. From prototype to usable system

This build is not a product. Getting there would need, roughly in this order: night operation, discussed below; approval to fly beyond visual line of sight, which is mainly a regulatory task rather than an engineering one; protection against water, dust and heat for monsoon conditions; and longer endurance than the present 15.3 minutes, which at this payload means a different type of airframe rather than a better battery.

For night operation, the obvious choice is not the best one. Thermal imaging is the usual answer and works well over rooftops, dry debris and open ground. It works poorly, though, for exactly the case this system is designed for. A person in floodwater cools towards the water's temperature, so the contrast a thermal camera relies on largely disappears, and what remains comes only from the head and shoulders, the smallest part of an already small target.

Near-infrared suits this case better. Water absorbs strongly above about 900 nm while skin reflects comparatively well, so a person appears bright against dark water. The same effect is why near-infrared is widely used to map water in remote sensing. It is also about ten times cheaper than a thermal camera, since it only needs a camera with its infrared filter removed, and it can be evaluated on public imagery before buying anything. The two would complement each other: thermal for survivors on dry ground at night, and near-infrared for survivors in water.

B. How little computing is enough?

Section IV-E shows that we cannot meet this task by shrinking the image, because for a fixed camera the size of a person in pixels and the area covered per pass trade against each other. A smaller detector input needs either a lower altitude or a longer lens, and both narrow the swath. What that argument does not tell us is where the limit actually lies.

Finding that limit would be useful in its own right. We could run the same tiled detection, on the same images, on three levels of hardware: the 26 TOPS accelerator used here, its 13 TOPS version, and a microcontroller-class device. For each we would record the altitude, tile count and detection rate at which recall drops below the requirement. The result would be a curve of achievable coverage rate against computing power. As far as we know, this has not been published for aerial person detection, and it is what a designer would need.

One application is flight on other planets, where power is severely limited. *Ingenuity* made 72 flights on Mars with a

TABLE X
PRINCIPAL RISKS AND MITIGATIONS

Risk	Sev.	Mitigation
Detection recall below target on real imagery	High	Higher because the field data campaign is deferred. The model is trained on public aerial datasets, and the tiling and frame-combining settings can be tuned without changing hardware
Detection and video encoding compete for one processor	Medium	Arises from the move to digital video. The companion has no hardware video encoder, so each aircraft encodes its own stream in software. Detection runs on the accelerator, and under load the video resolution and frame rate drop, never the tile count. P5 measures processor headroom with both running
Heading depends on a compass near high-current wiring	Medium	Moving-baseline heading is possible with our receiver but deferred for mass and cost, so heading comes from the compass in the rover unit, costing 18 mm in the case-C budget. P6 measures heading accuracy, and we fit a second receiver if it is worse than 1°
Flight-test window shortened by delivery delays	High	Higher because the training airframe and most spares are deferred. Our only protection is lead time, so long-lead items are ordered in the first tranche
Recovery parachute deferred	Medium	The 300 g allowance is kept in the sizing but no parachute is bought, so nothing limits the damage from a crash. We have not found an Indian supplier; mass rating and lead time would need checking if we reverse the deferral
Setup takes longer than allowed	Medium	Modelled, not measured. Timed setup drills are our highest-priority early test
Airframe overhangs the pad edge on touchdown	Medium	Clearance is exactly one airframe radius, so any touchdown error uses up the margin. We will measure touchdown error in P6 before accepting the slot layout
Motor thrust below what we need	High	The motors have no published thrust curve, and we cannot test them before they arrive. The heavier LiPo pack makes this worse: keeping thrust-to-weight at 2.0 at 7.1 kg needs 3.55 kgf per motor. We measure thrust on arrival; a shortfall would force a propulsion change after the flight-test window has started
Airframe lost during testing	High	Higher because spares are limited. We keep a set of structural parts and consumables, but a second loss in the same week would not be covered
Flight insurance may be compulsory	Medium	Deferred by team decision. To be checked against the rulebook and DGCA rules before any flight

1.8 kg helicopter in an atmosphere less than one per cent as dense as Earth's, and *Dragonfly* is due to launch for Titan in 2028. In both cases commands take minutes to arrive, so all perception decisions have to be made on board. Microcontroller-class detection is often proposed for such missions, and our measurement would show what target size that choice gives up.

C. Cost in larger numbers

Our part prices are single-unit retail prices for a build of three. Each extra aircraft would already cost noticeably less than our per-aircraft figure, because the ground station, test equipment and spares are bought only once. At production volumes the important factors would be different, such as tooling, negotiated component prices and assembly labour, and we cannot estimate these reliably from a three-aircraft bill of materials, so we do not try.

D. Possible commercial use

The likely customers are state disaster management authorities and the national and state disaster response forces. To judge whether this could be a viable business, we would need three things this project does not attempt: how many such systems these agencies would buy and how often, what they currently pay for similar capability, and the hourly operating cost of the boat or helicopter searches it would replace. These questions need published tender data and interviews with operators. We leave them open rather than guess.

One useful by-product of this project is a part-by-part record of which UAV components we could source in India and which we could not. That record will be useful whatever happens after this build.

XIII. CONCLUSION

We have described a system of three autonomous aircraft intended to speed up the slowest part of post-flood rescue, which is finding survivors. Throughout the work we have tried to keep what has been demonstrated separate from what has only been designed. The items left out of this build were deferred to save cost, and Section VIII sets out what each one costs the system.

The design point and error budgets are complete and can be reproduced. Coordination and failsafe behaviour have been shown in simulation with three autopilots running at once. What remains is to build and fly the aircraft and replace the modelled and assumed figures in Section VI-B with measured ones.

Our claim for the work so far is a modest one: every figure in this paper can be checked against the model that produced it, and every figure that has not yet been measured is marked as such.

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